

BFS Traversal

Step 1: Start at A

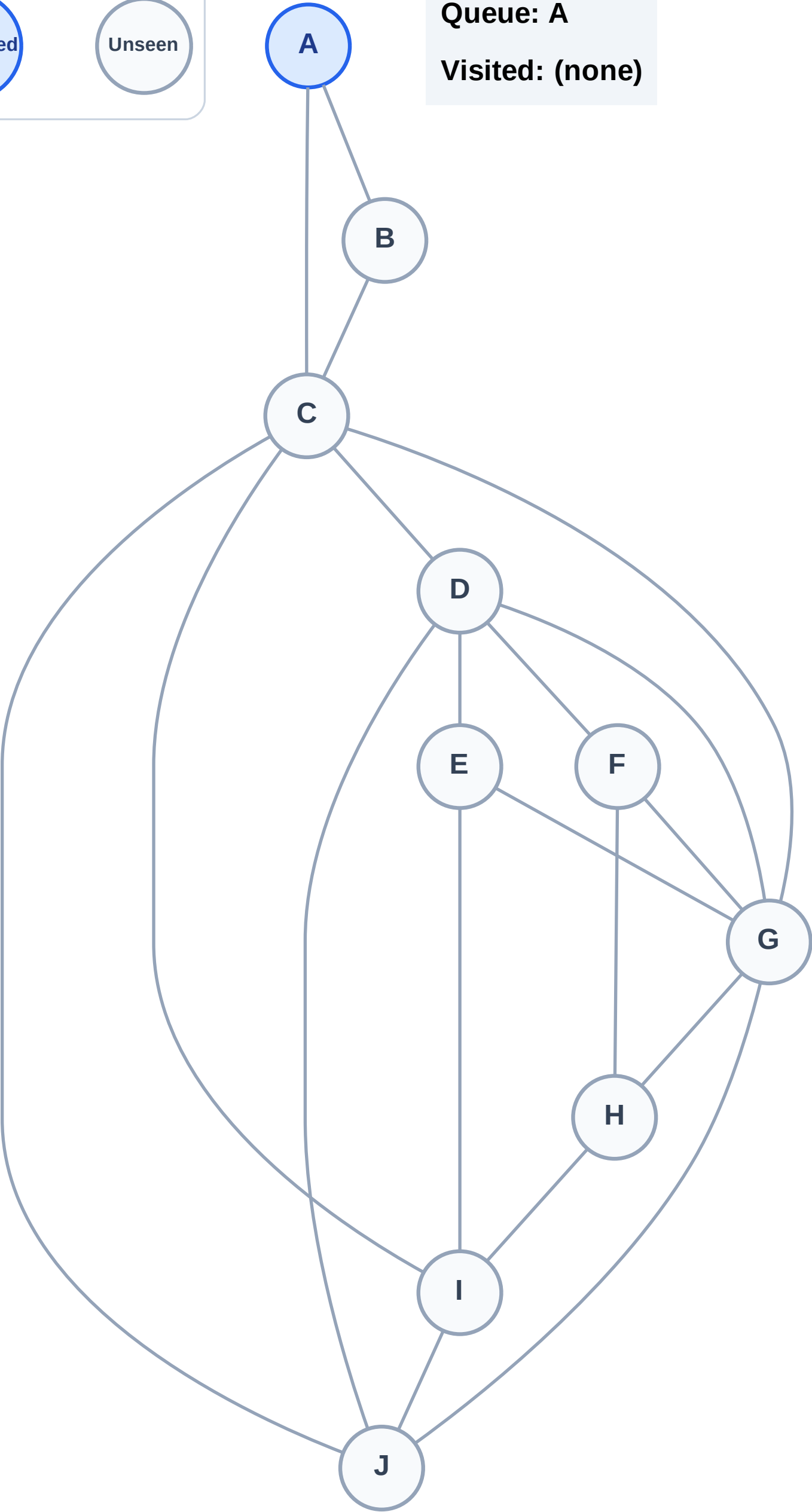
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

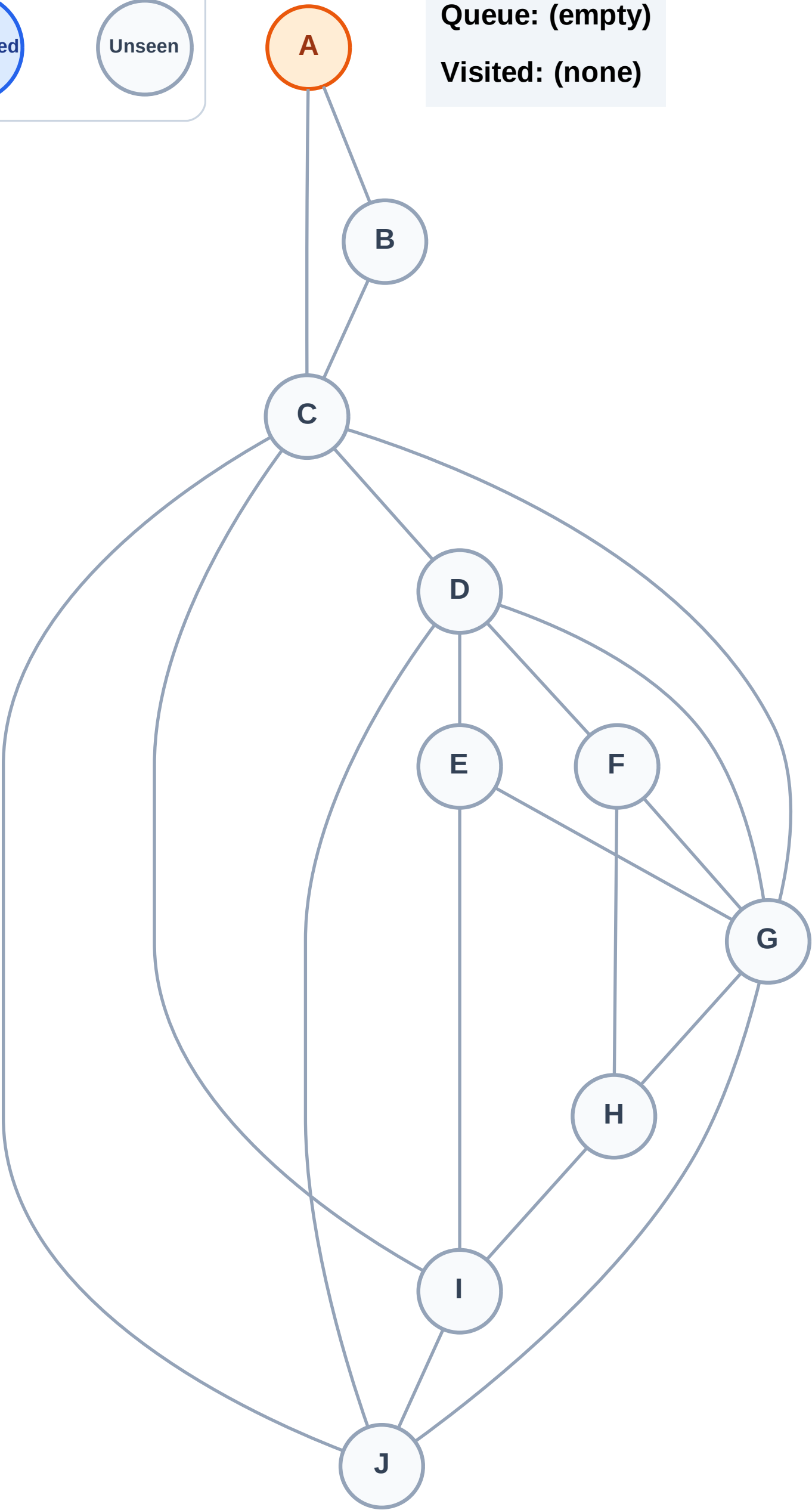
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C from A

Legend

Complete

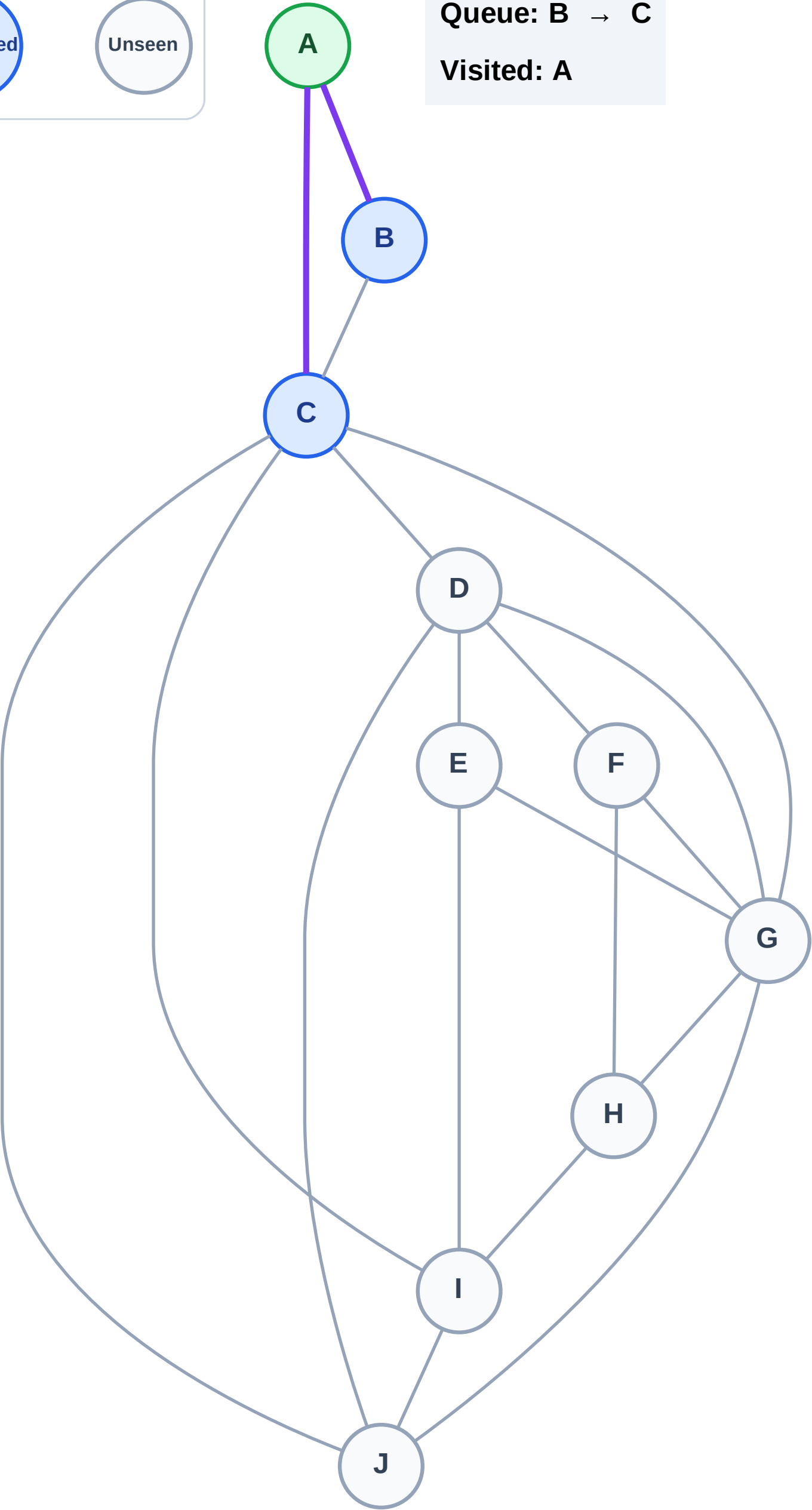
Processing

Discovered

Unseen

Queue: B → C

Visited: A



BFS Traversal

Step 4: Process node B

Legend

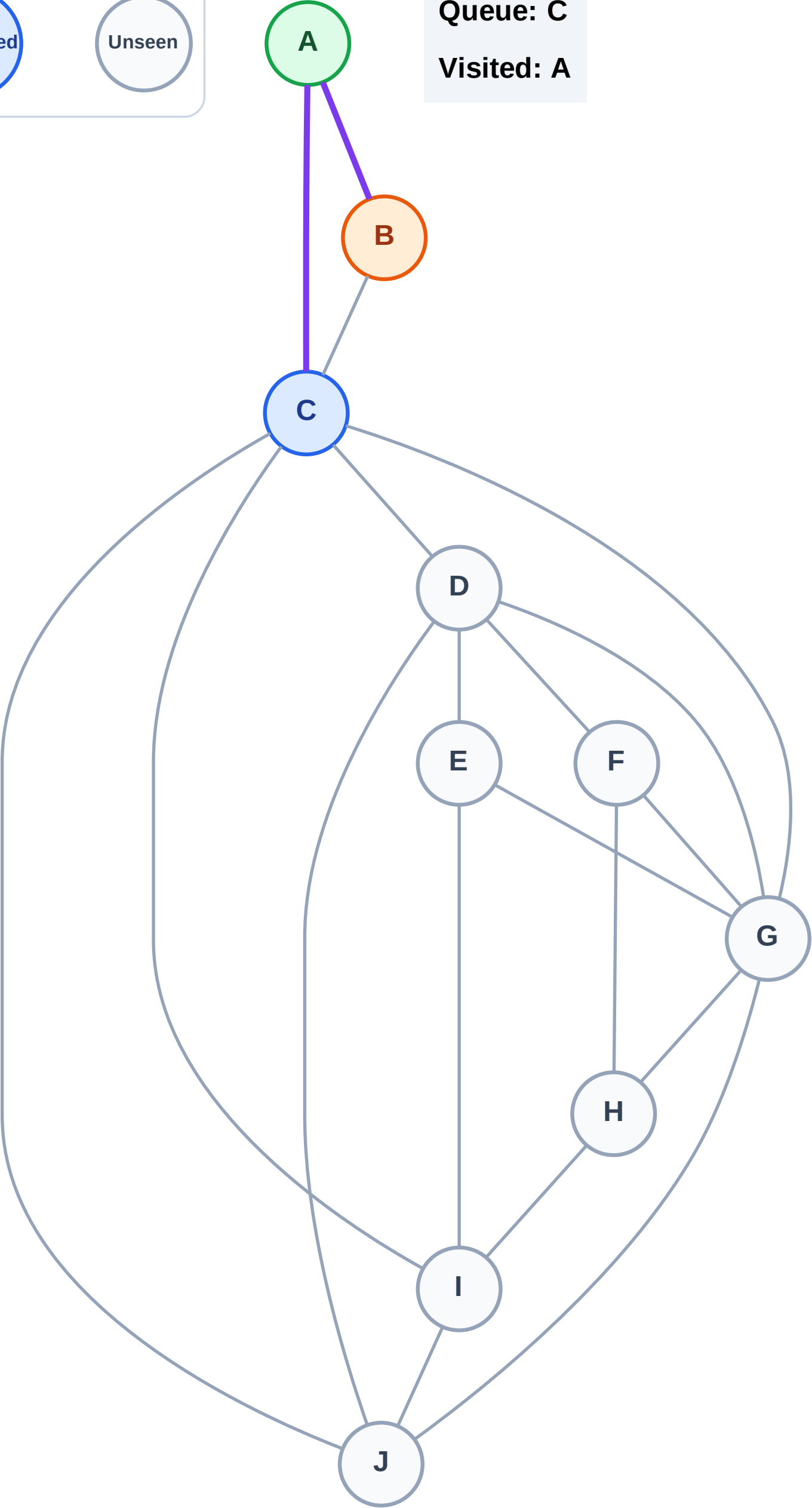
Complete

Processing

Discovered

Unseen

Queue: C
Visited: A



BFS Traversal

Step 5: No new neighbors from B

Complete

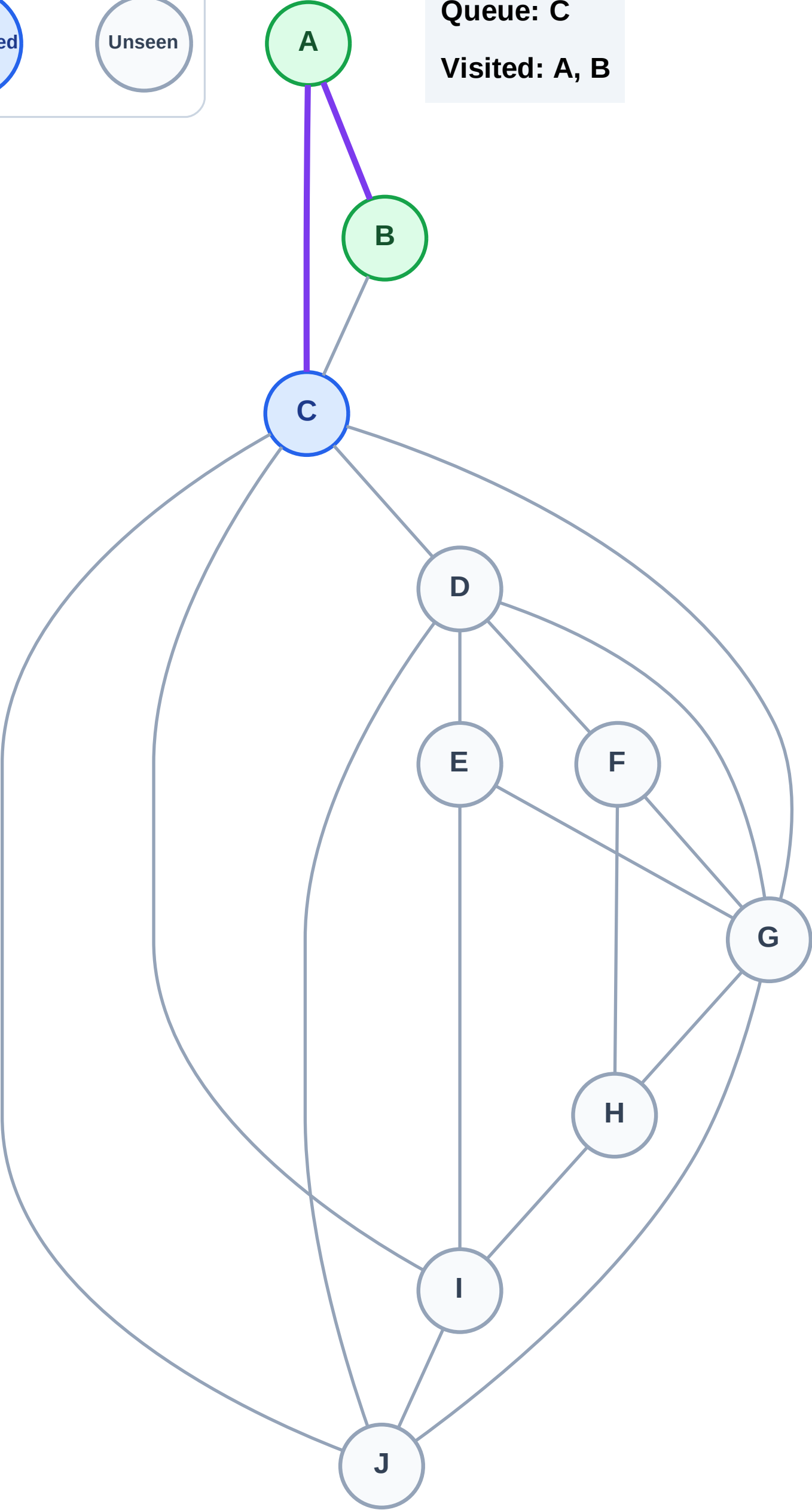
Processing

Discovered

Unseen

Queue: C

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

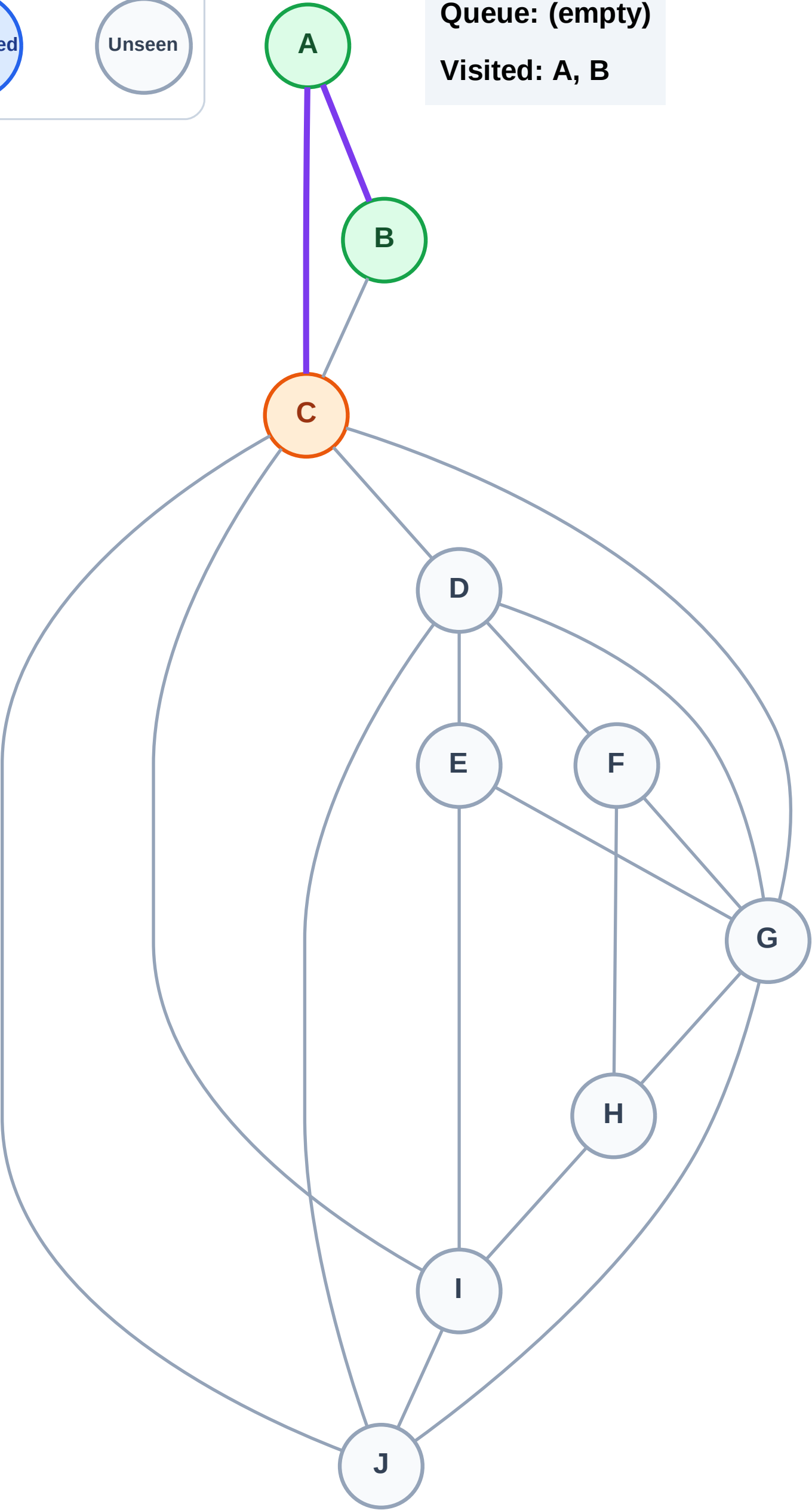
Complete

Processing

Discovered

Unseen

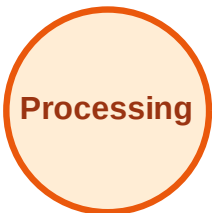
Queue: (empty)
Visited: A, B



BFS Traversal

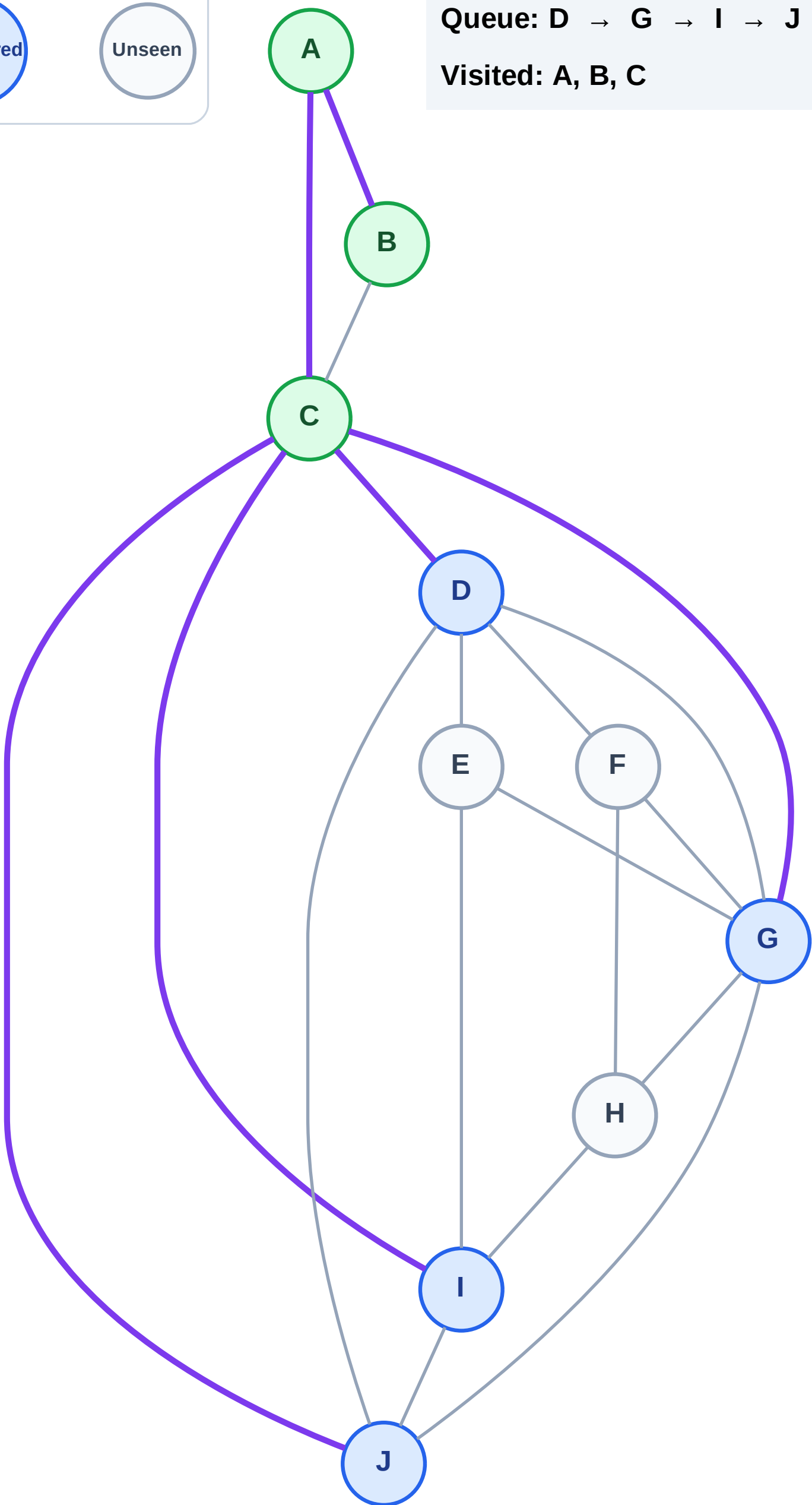
Step 7: Discover D, G, I, J from C

Legend



Queue: D → G → I → J

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

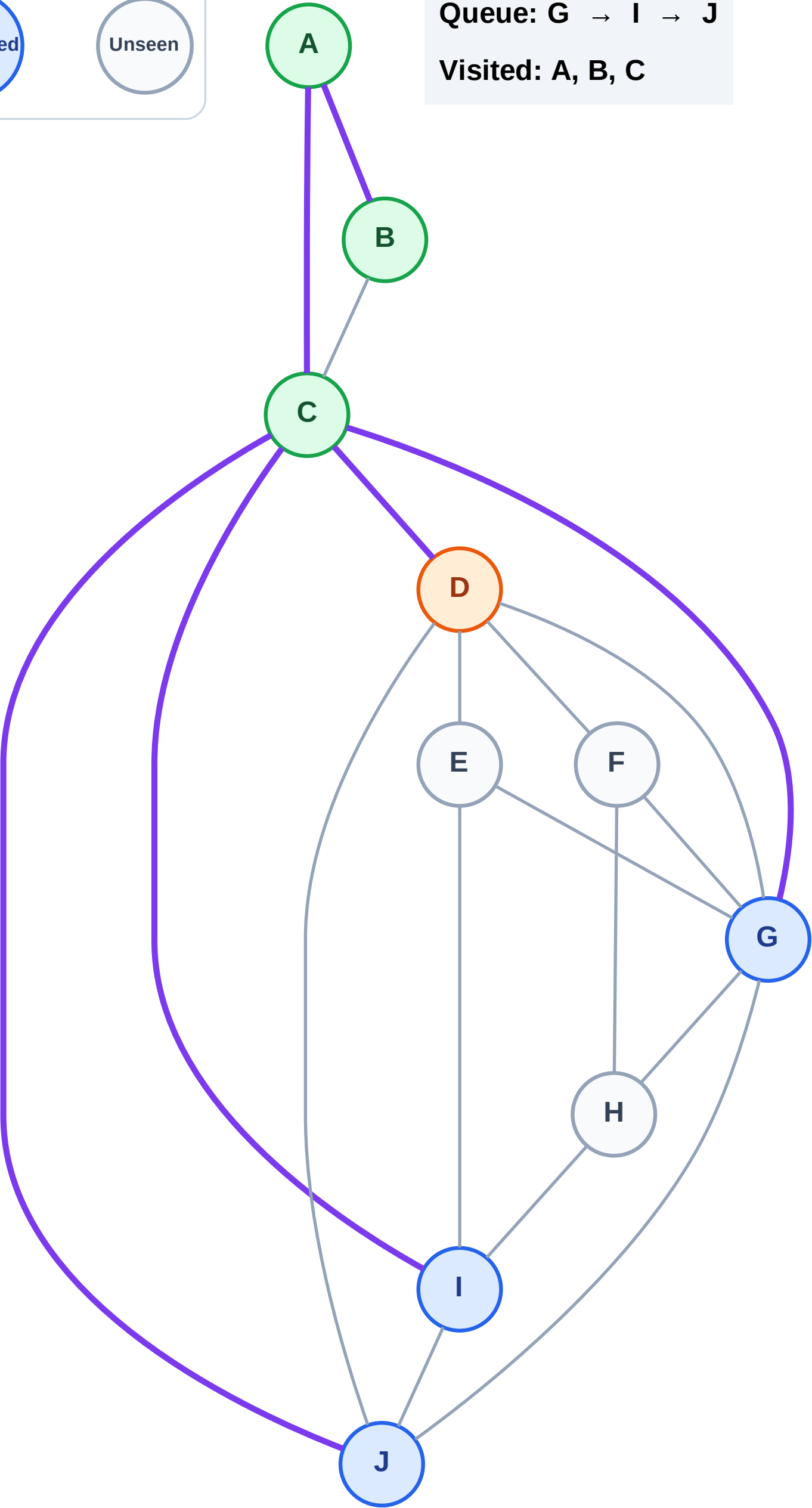
Processing

Discovered

Unseen

Queue: G → I → J

Visited: A, B, C



BFS Traversal

Step 9: Discover E, F from D

Legend

Complete

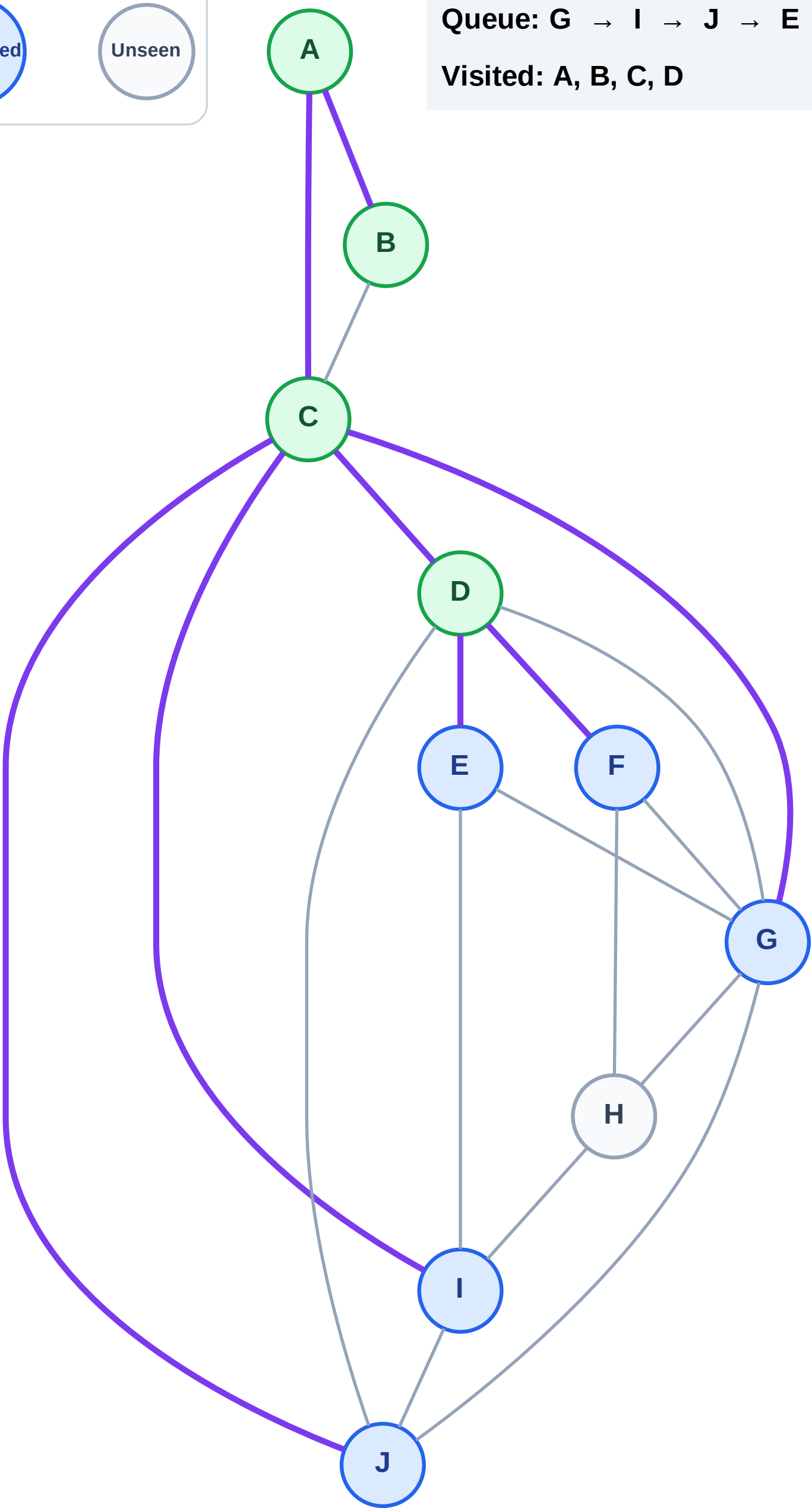
Processing

Discovered

Unseen

Queue: G → I → J → E → F

Visited: A, B, C, D



BFS Traversal

Step 10: Process node G

Legend

Complete

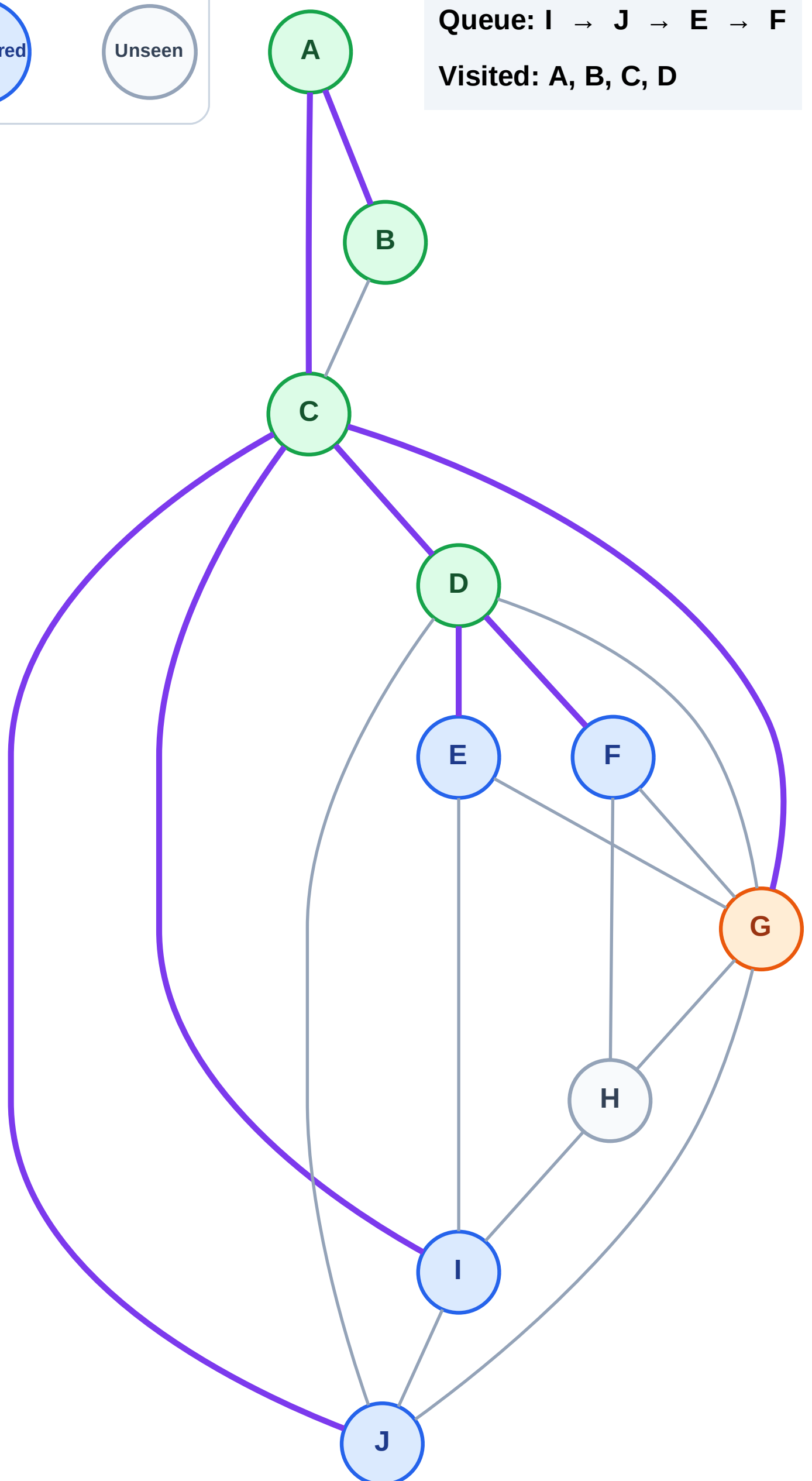
Processing

Discovered

Unseen

Queue: I → J → E → F

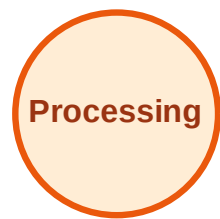
Visited: A, B, C, D



BFS Traversal

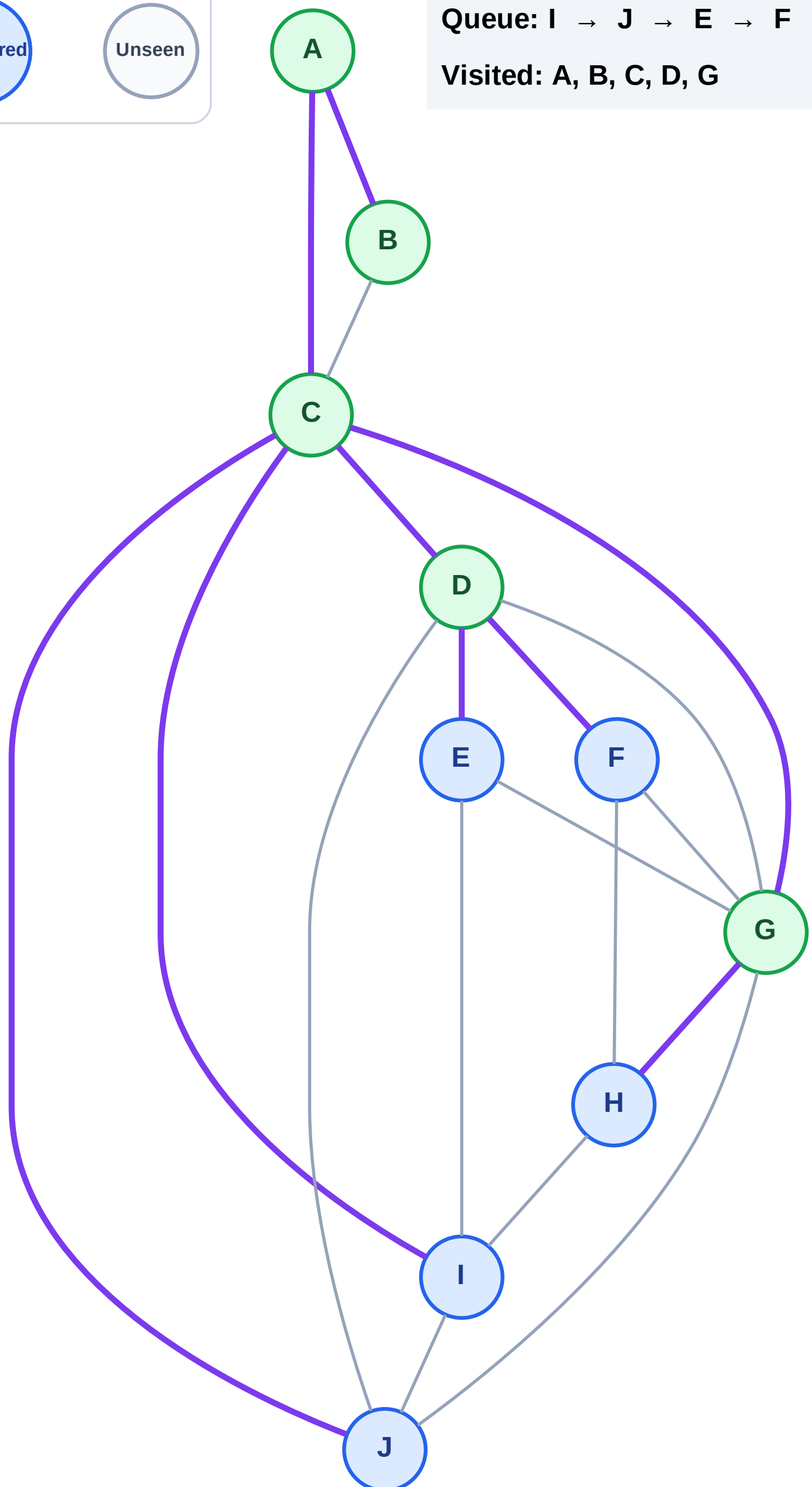
Step 11: Discover H from G

Legend



Queue: I → J → E → F → H

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node I

Legend

Complete

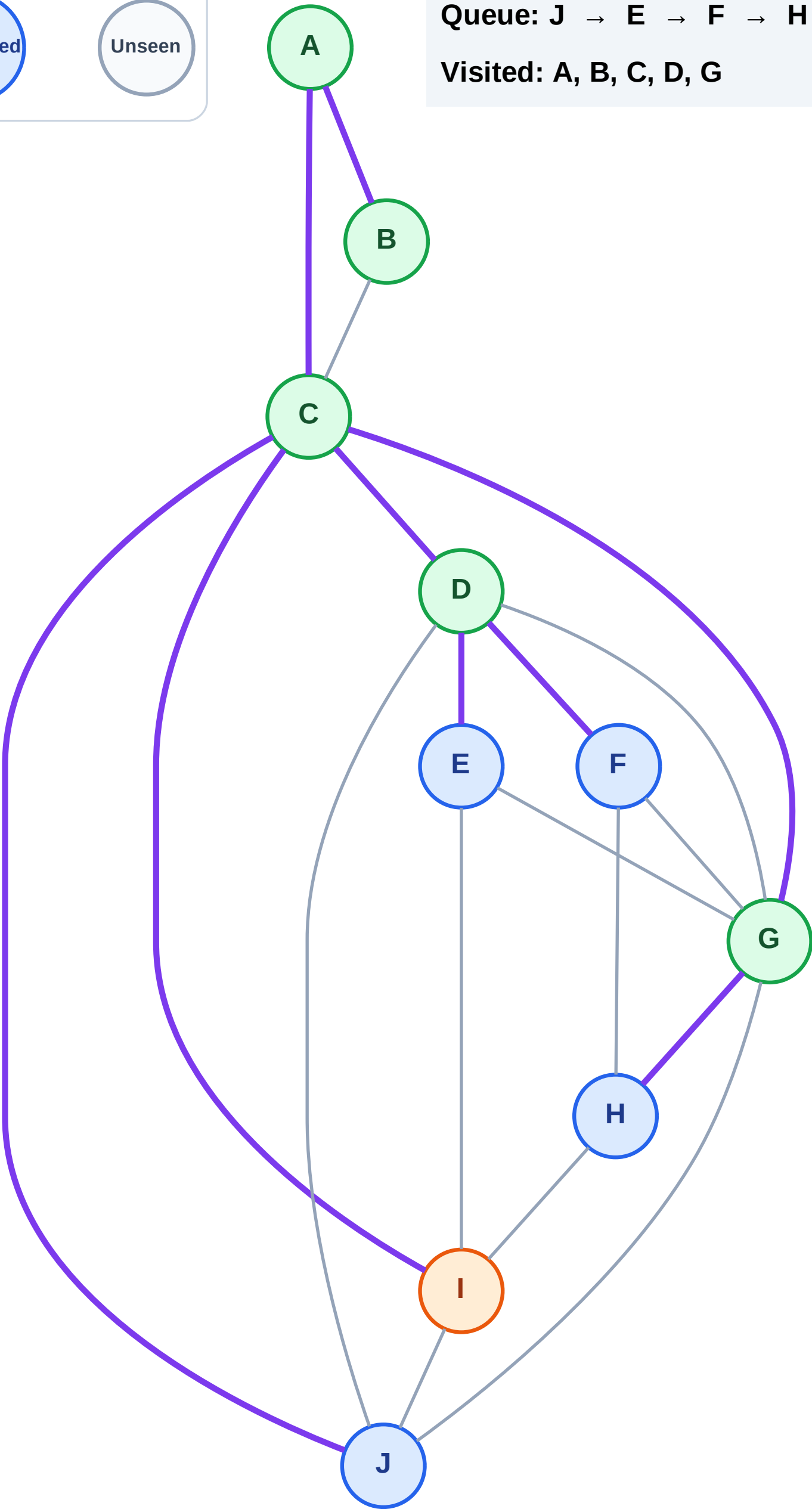
Processing

Discovered

Unseen

Queue: J → E → F → H

Visited: A, B, C, D, G



BFS Traversal

Step 13: No new neighbors from I

Legend

Complete

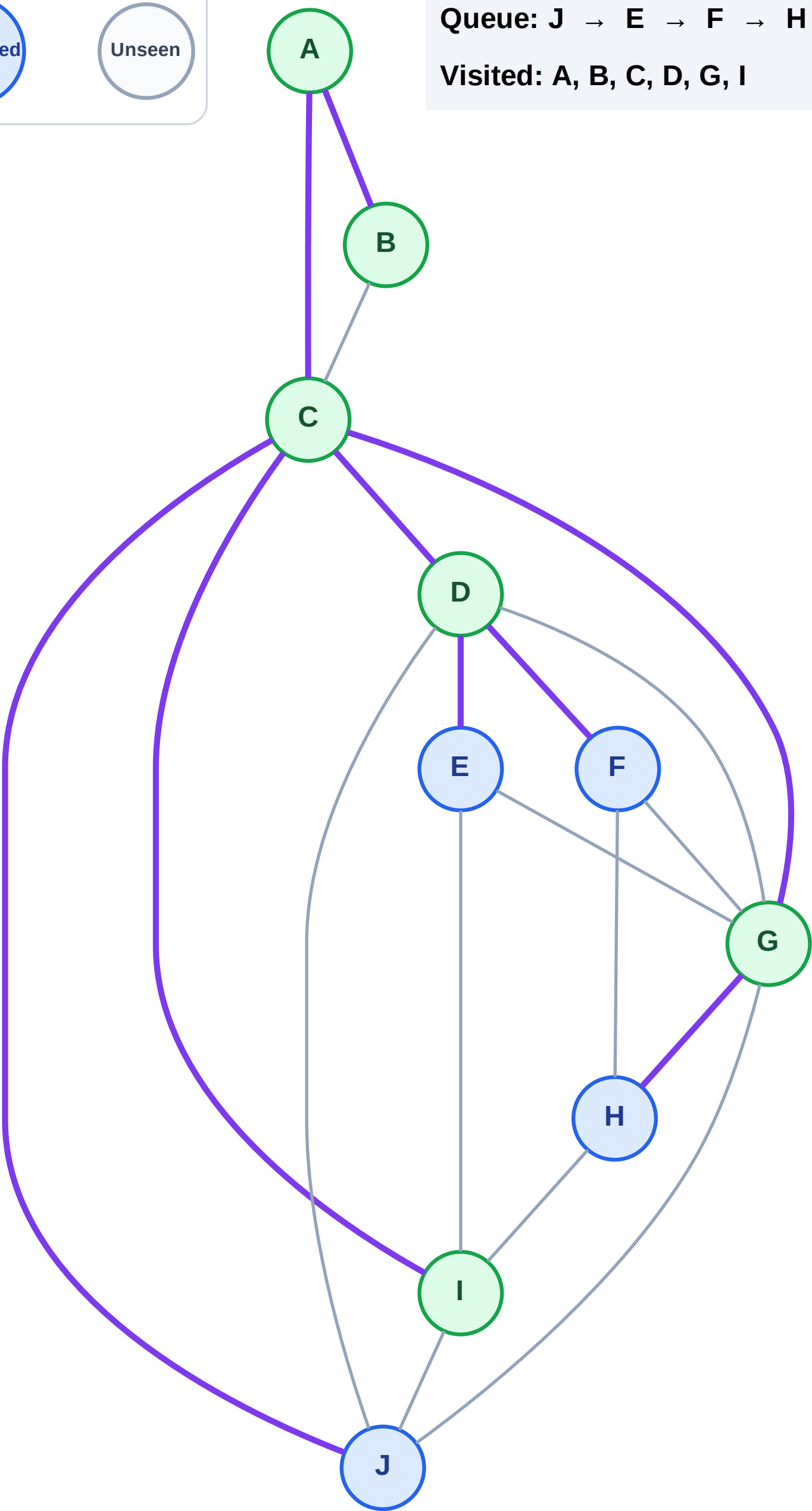
Processing

Discovered

Unseen

Queue: J → E → F → H

Visited: A, B, C, D, G, I



BFS Traversal

Step 14: Process node J

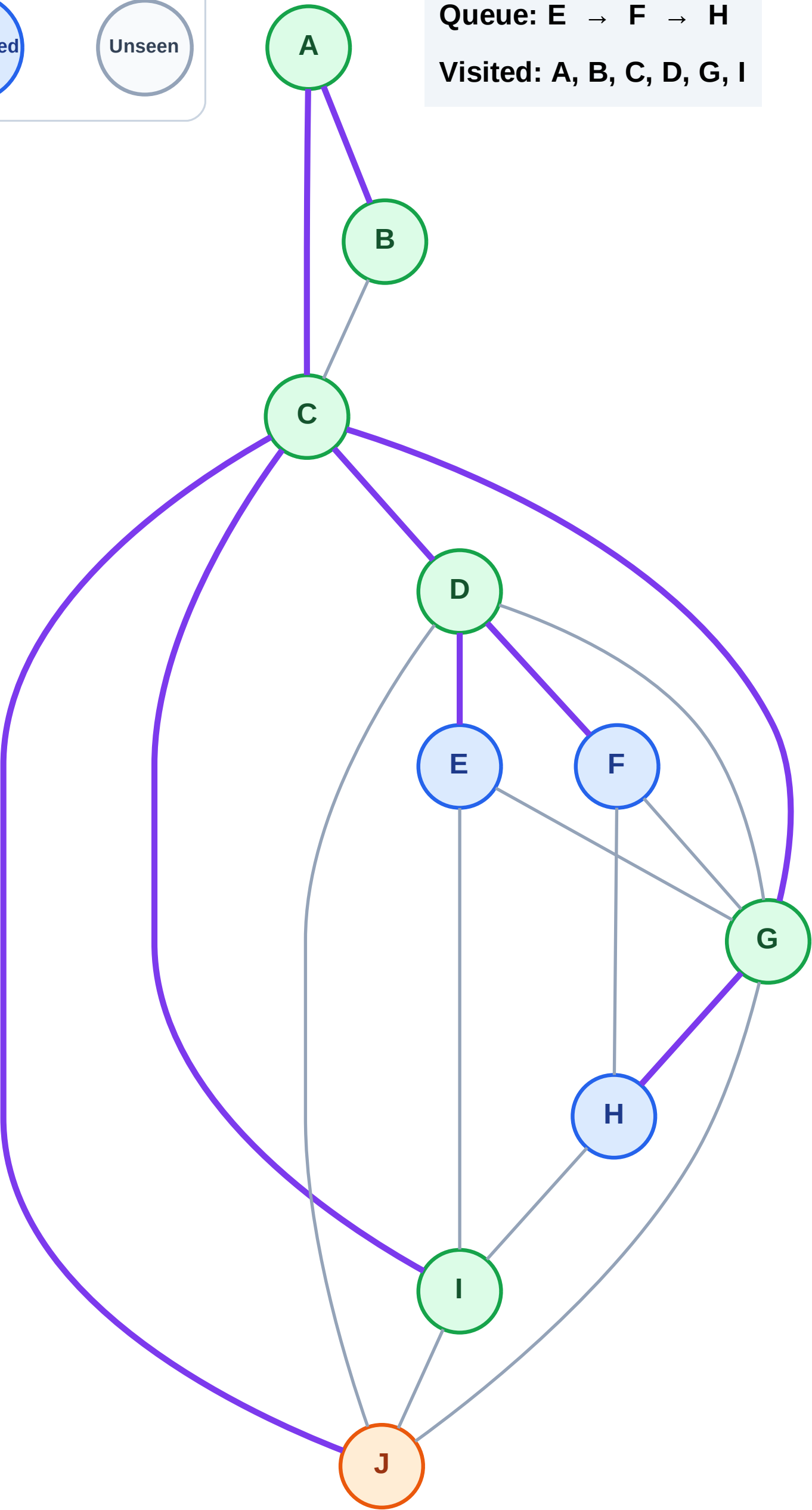
Complete

Processing

Discovered

Unseen

Queue: E → F → H
Visited: A, B, C, D, G, I



BFS Traversal

Step 15: No new neighbors from J

Legend

Complete

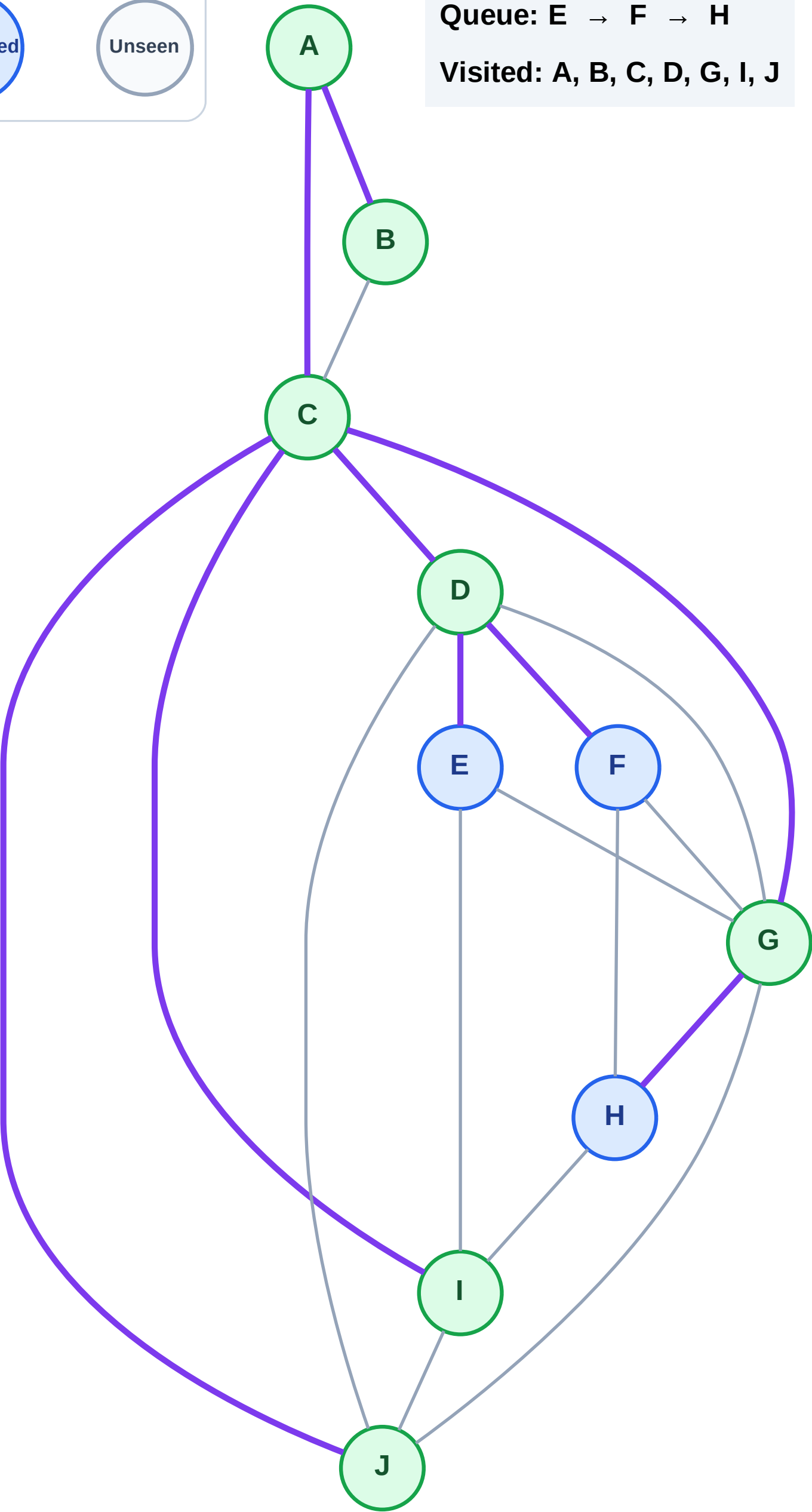
Processing

Discovered

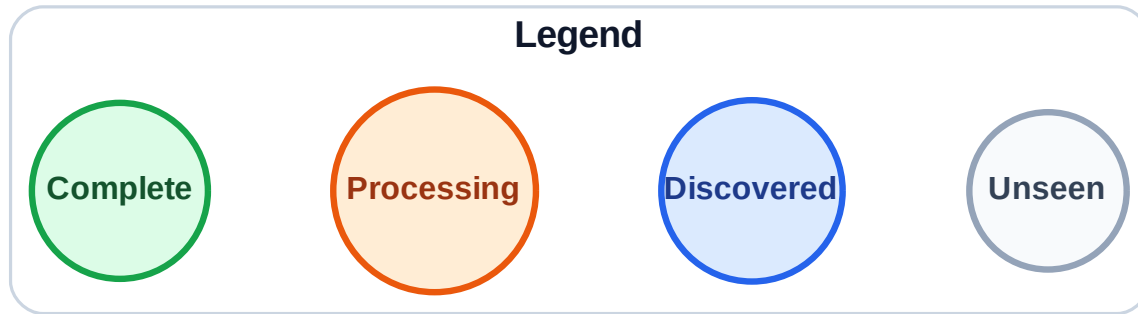
Unseen

Queue: E → F → H

Visited: A, B, C, D, G, I, J

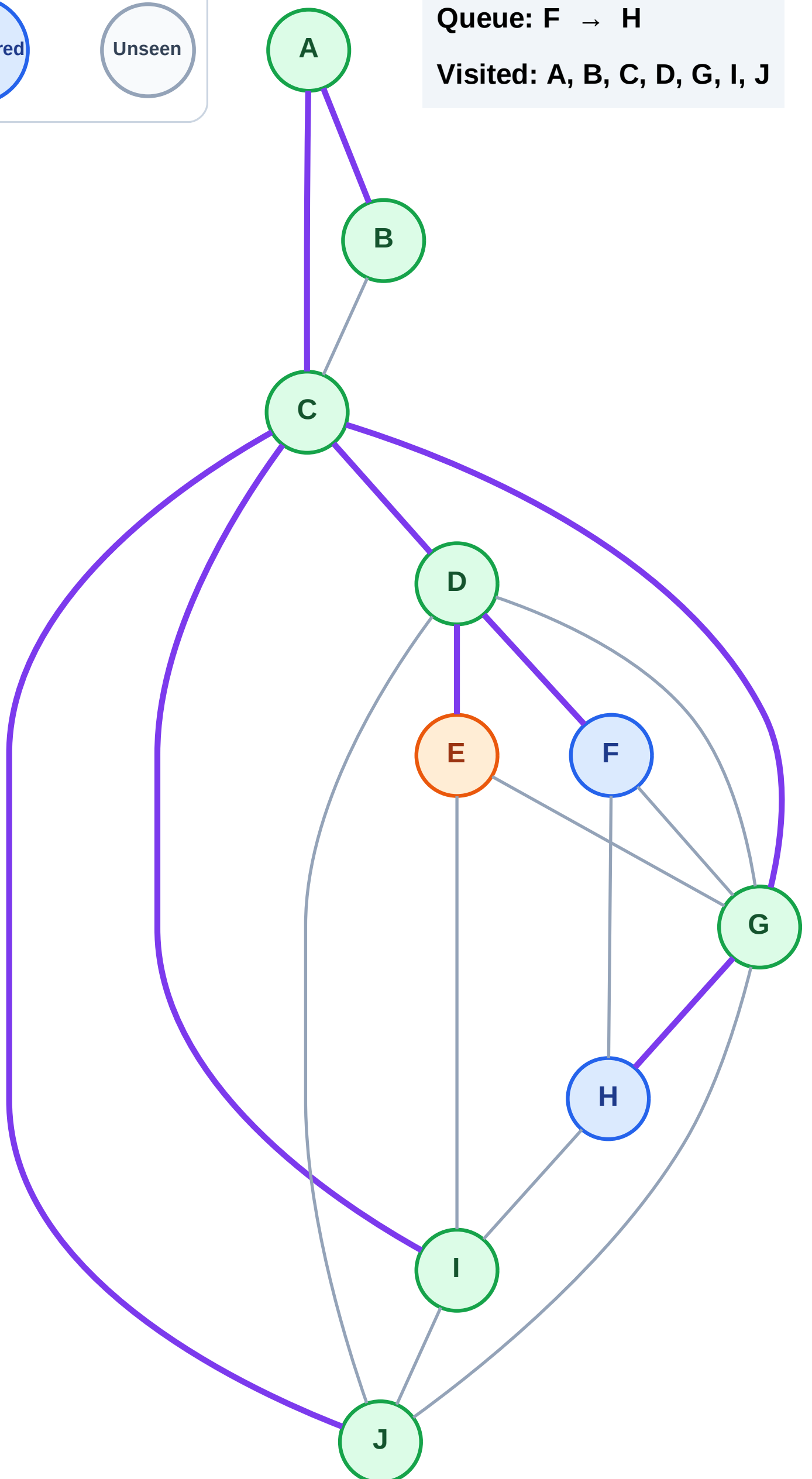


Step 16: Process node E



Queue: F → H

Visited: A, B, C, D, G, I, J



BFS Traversal

Step 17: No new neighbors from E

Legend

Complete

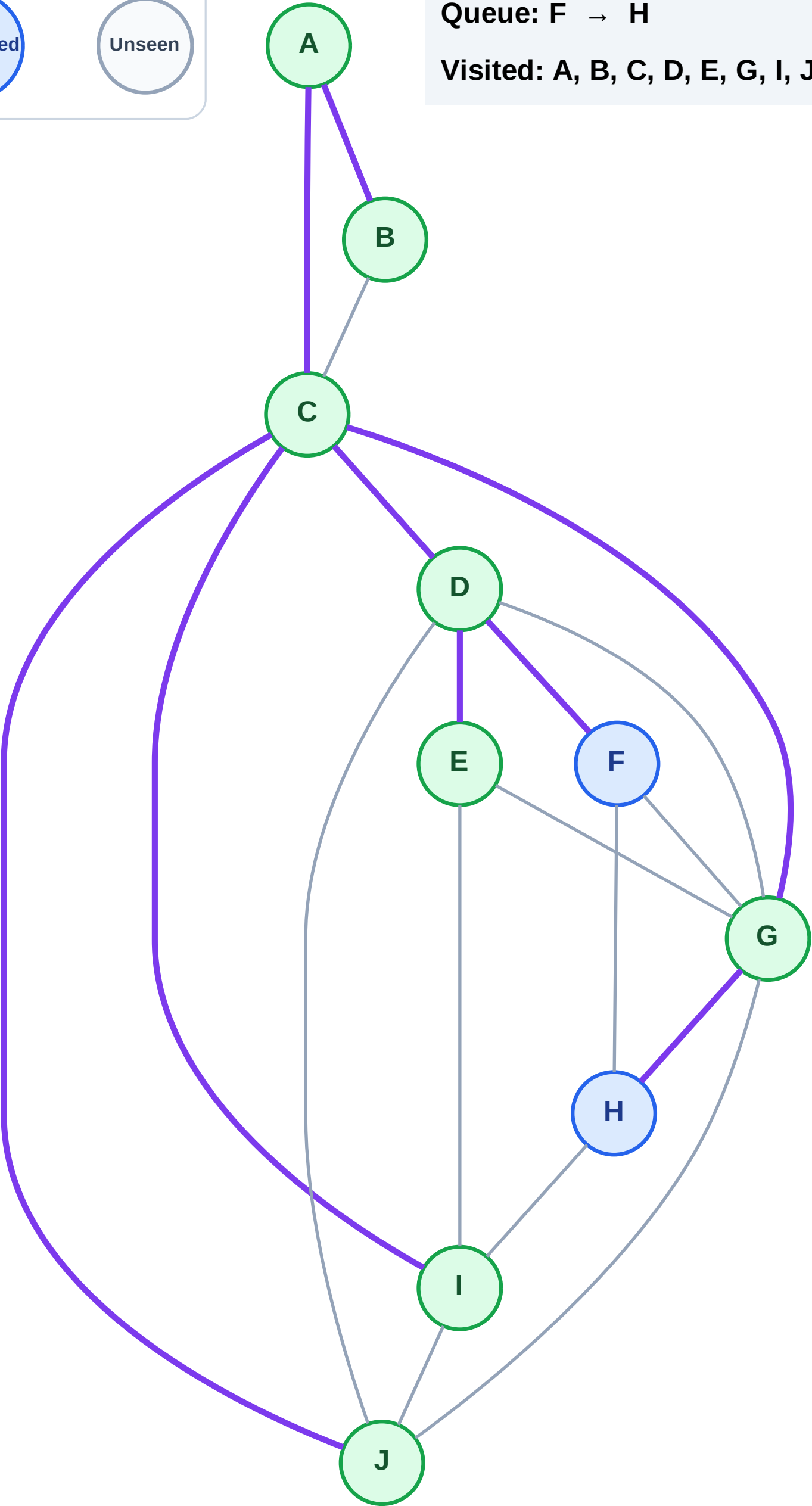
Processing

Discovered

Unseen

Queue: F → H

Visited: A, B, C, D, E, G, I, J



BFS Traversal

Step 18: Process node F

Legend

Complete

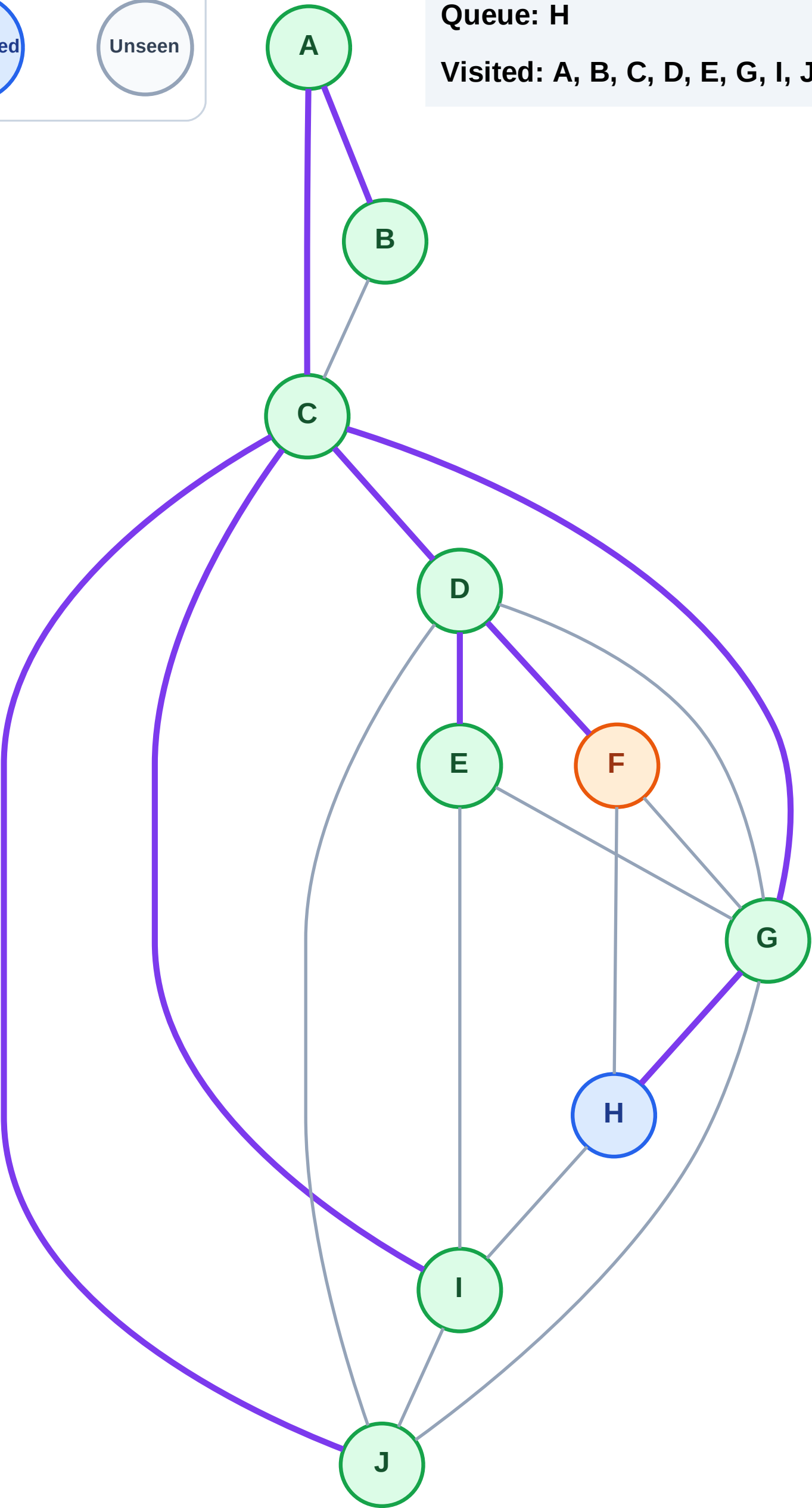
Processing

Discovered

Unseen

Queue: H

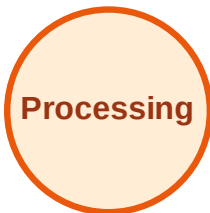
Visited: A, B, C, D, E, G, I, J



BFS Traversal

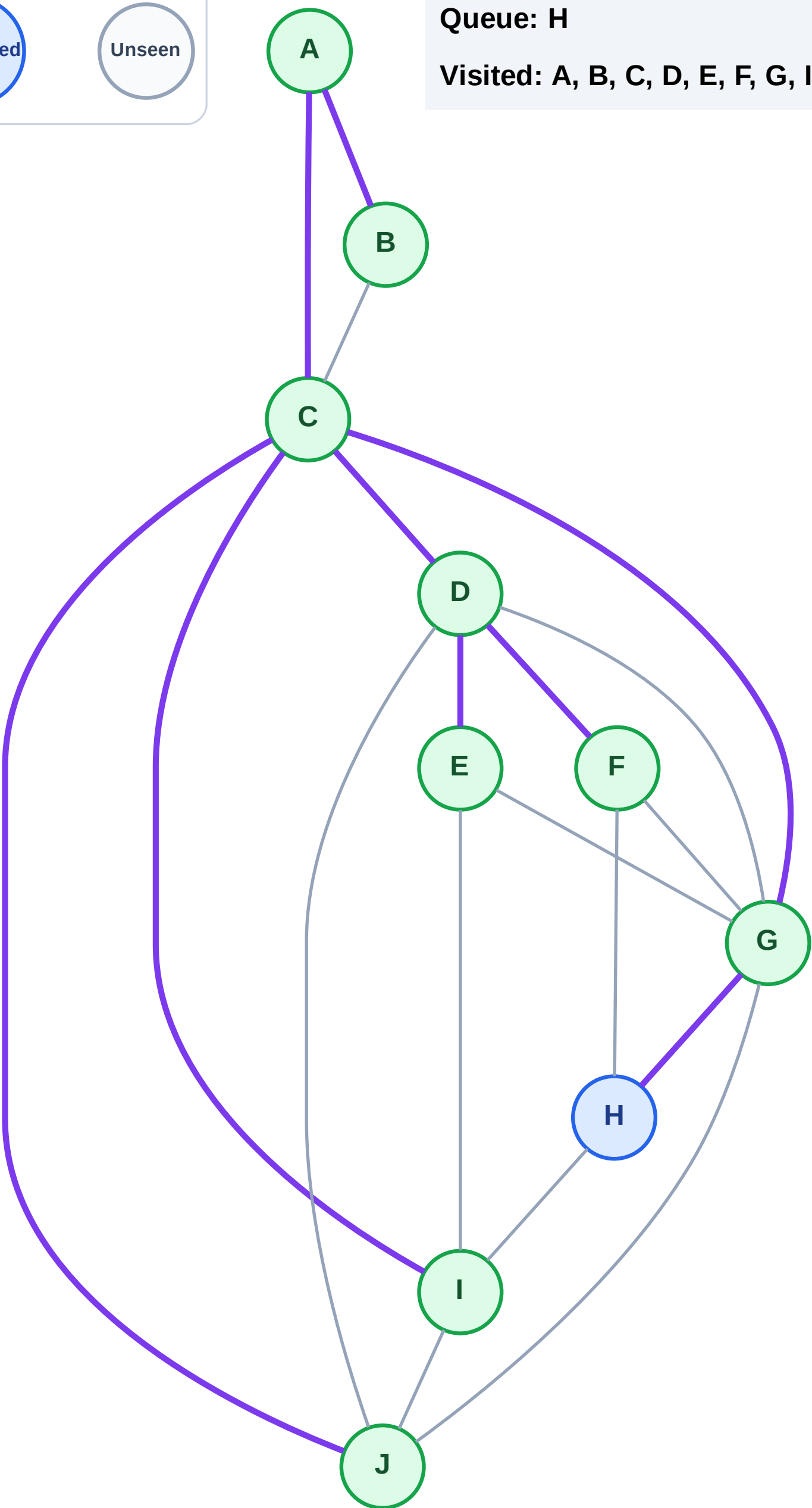
Step 19: No new neighbors from F

Legend



Queue: H

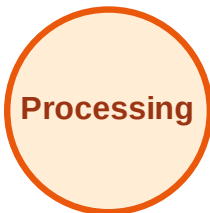
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

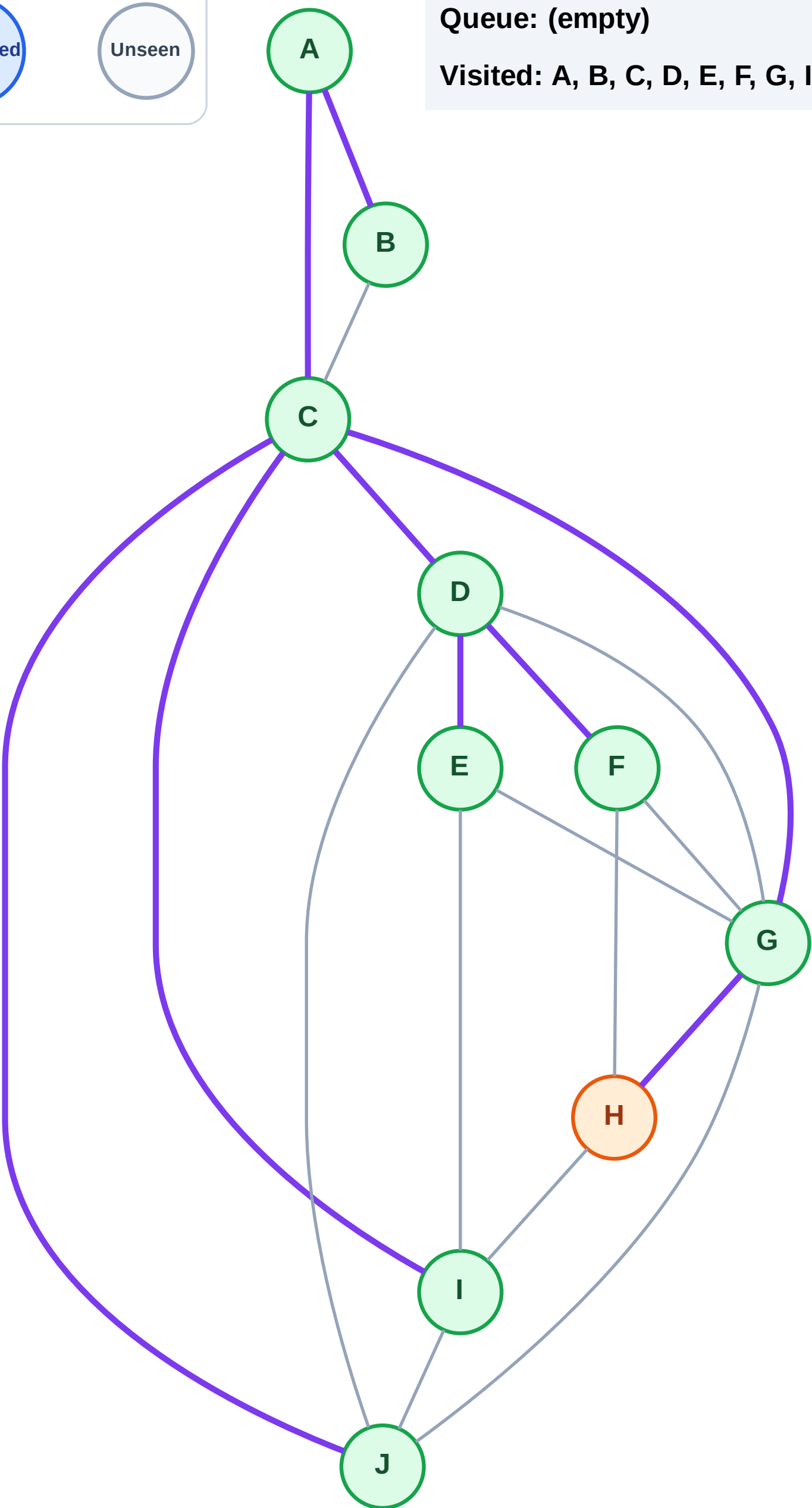
Step 20: Process node H

Legend



Queue: (empty)

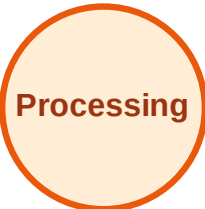
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

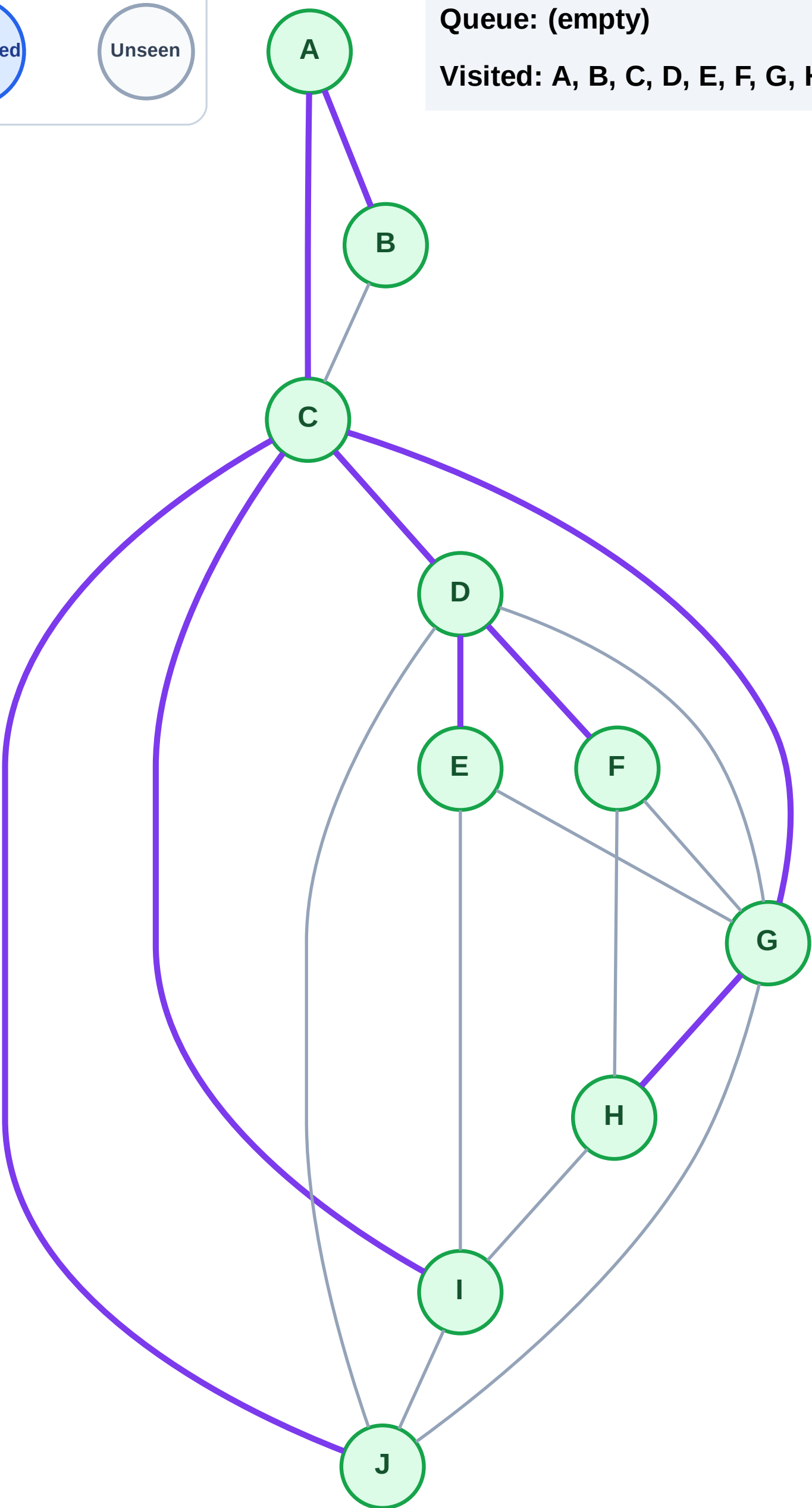
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

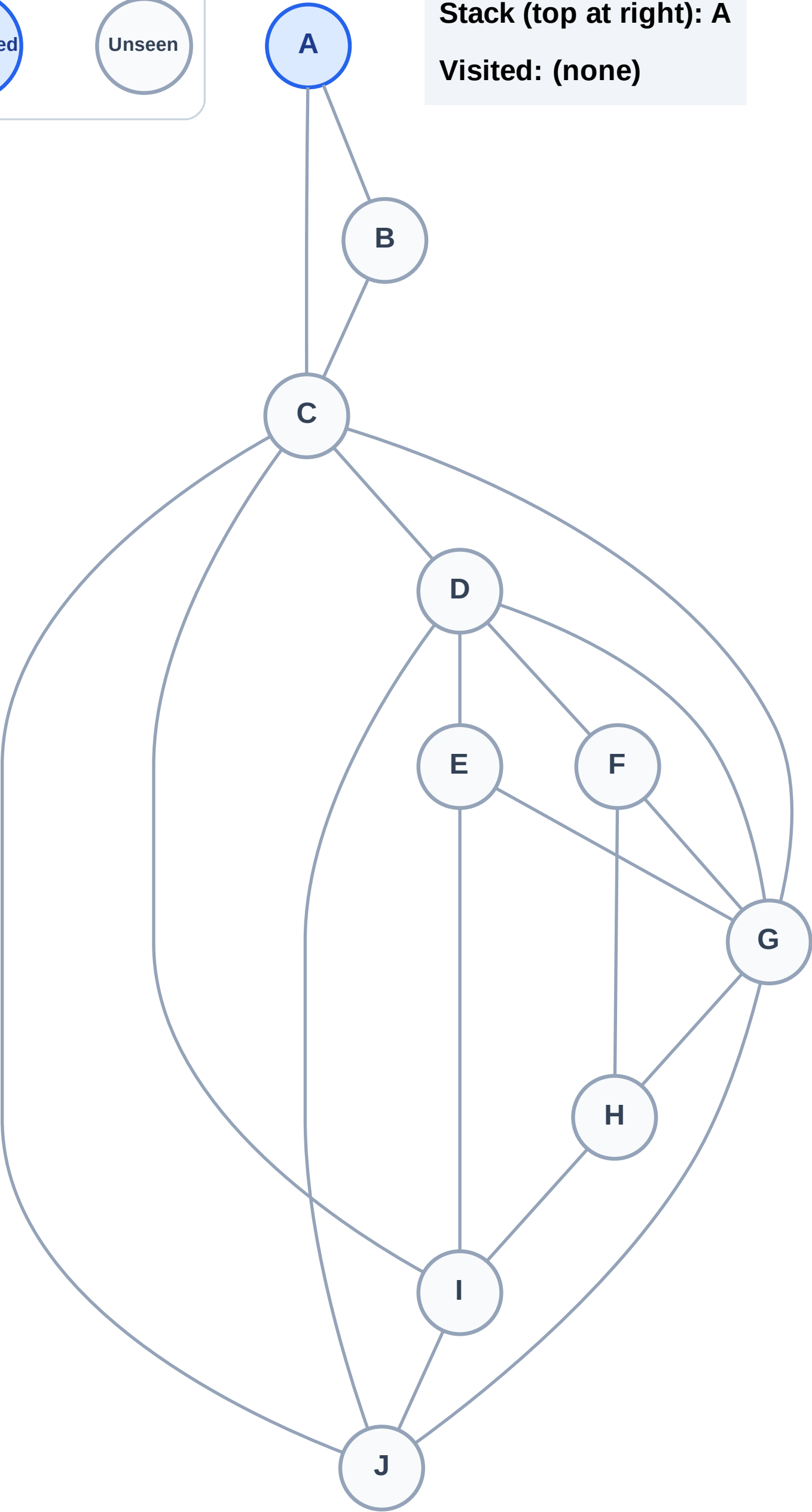
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

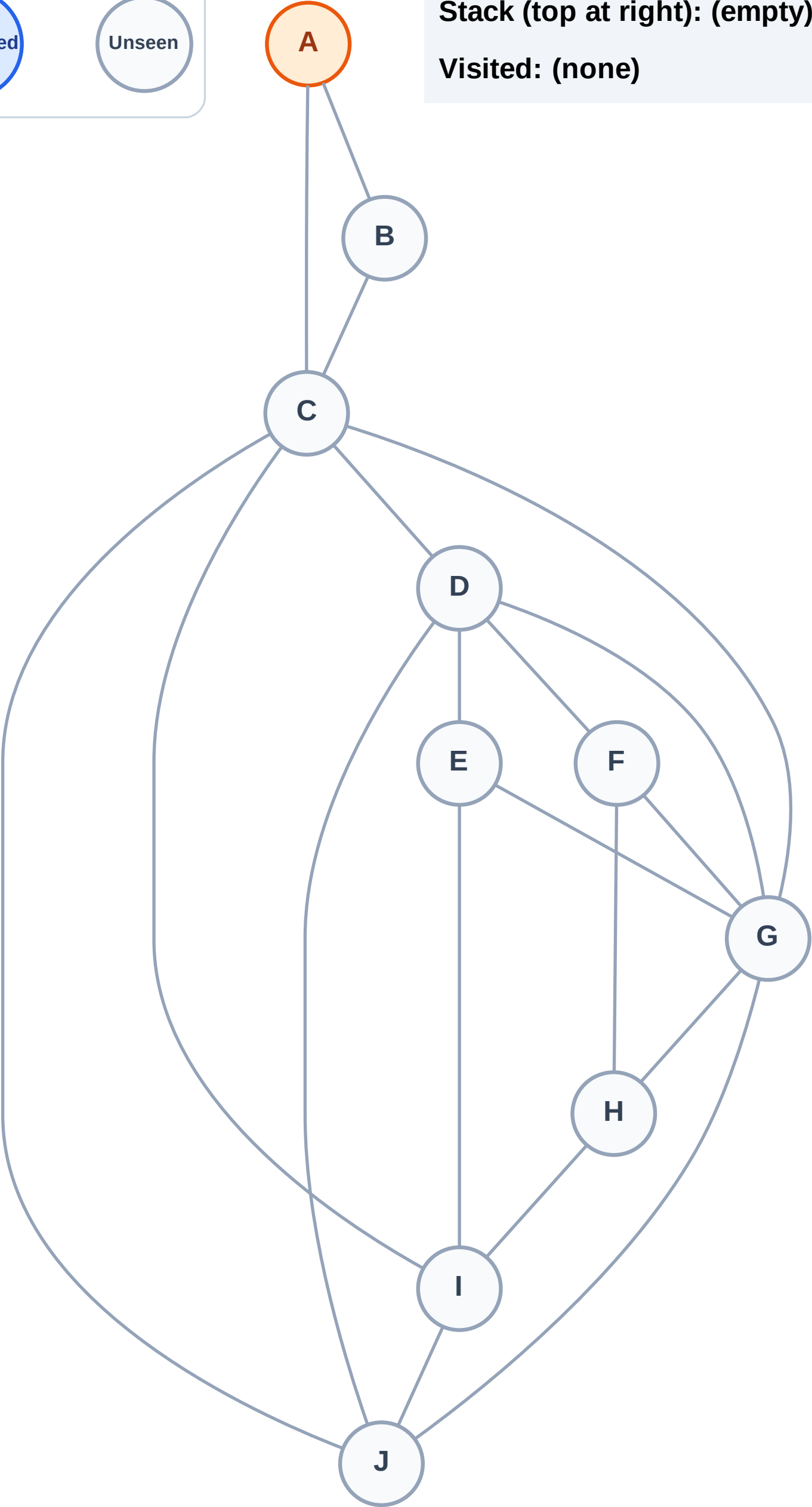
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover C, B from A

Legend

Complete

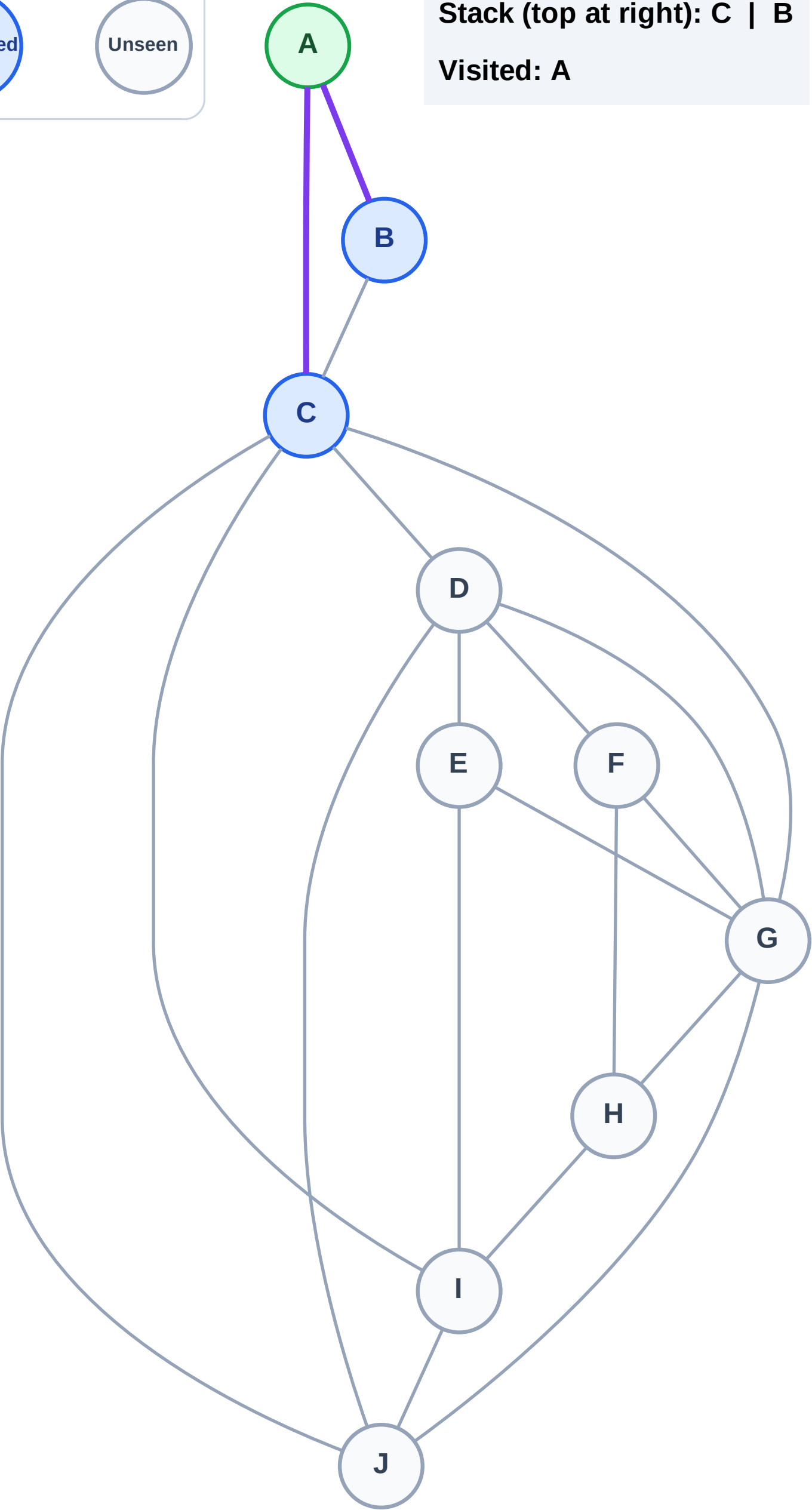
Processing

Discovered

Unseen

Stack (top at right): C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

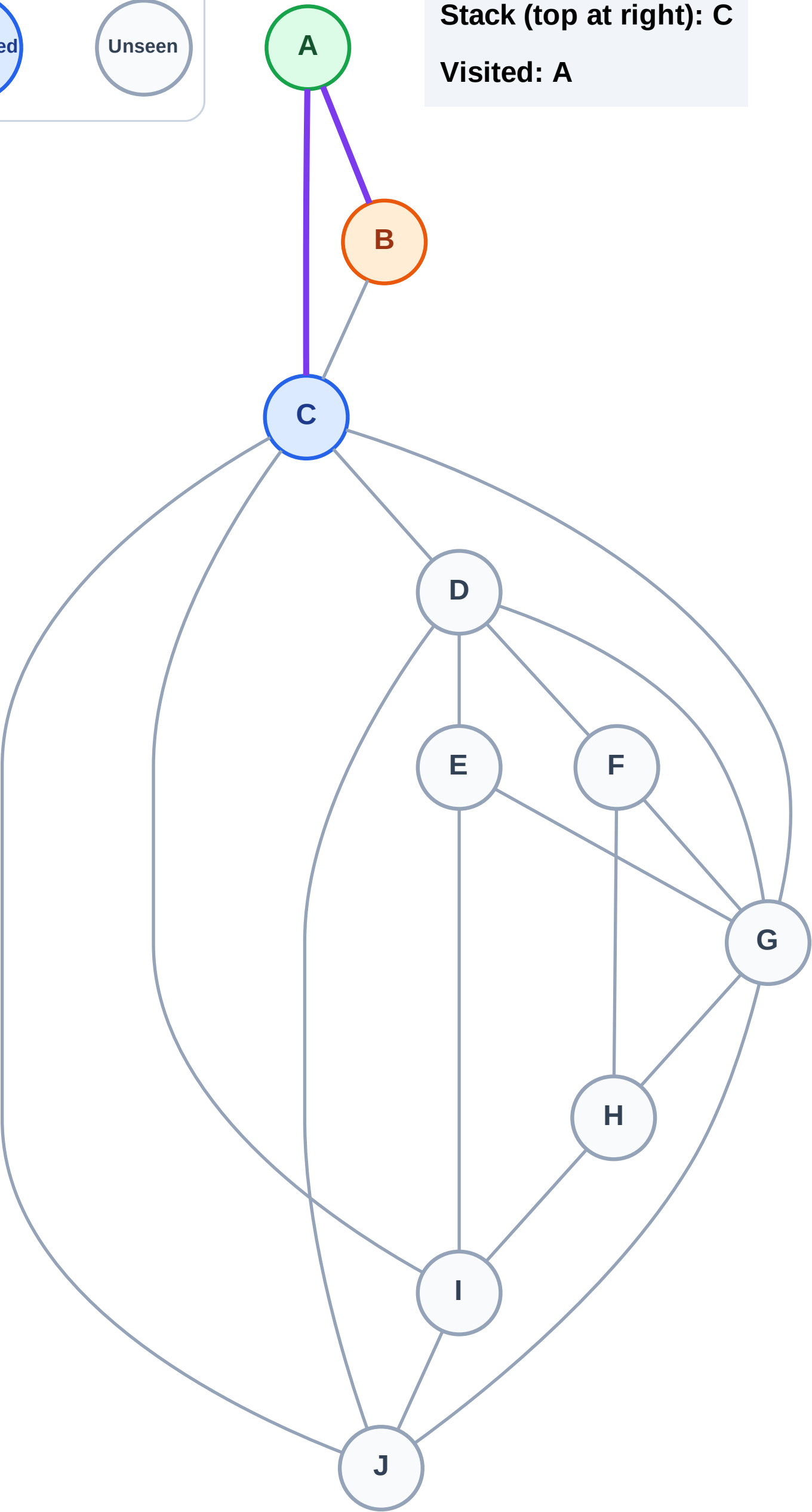
Complete

Processing

Discovered

Unseen

Stack (top at right): C
Visited: A



DFS Traversal

Step 5: No new neighbors from B

Legend

Complete

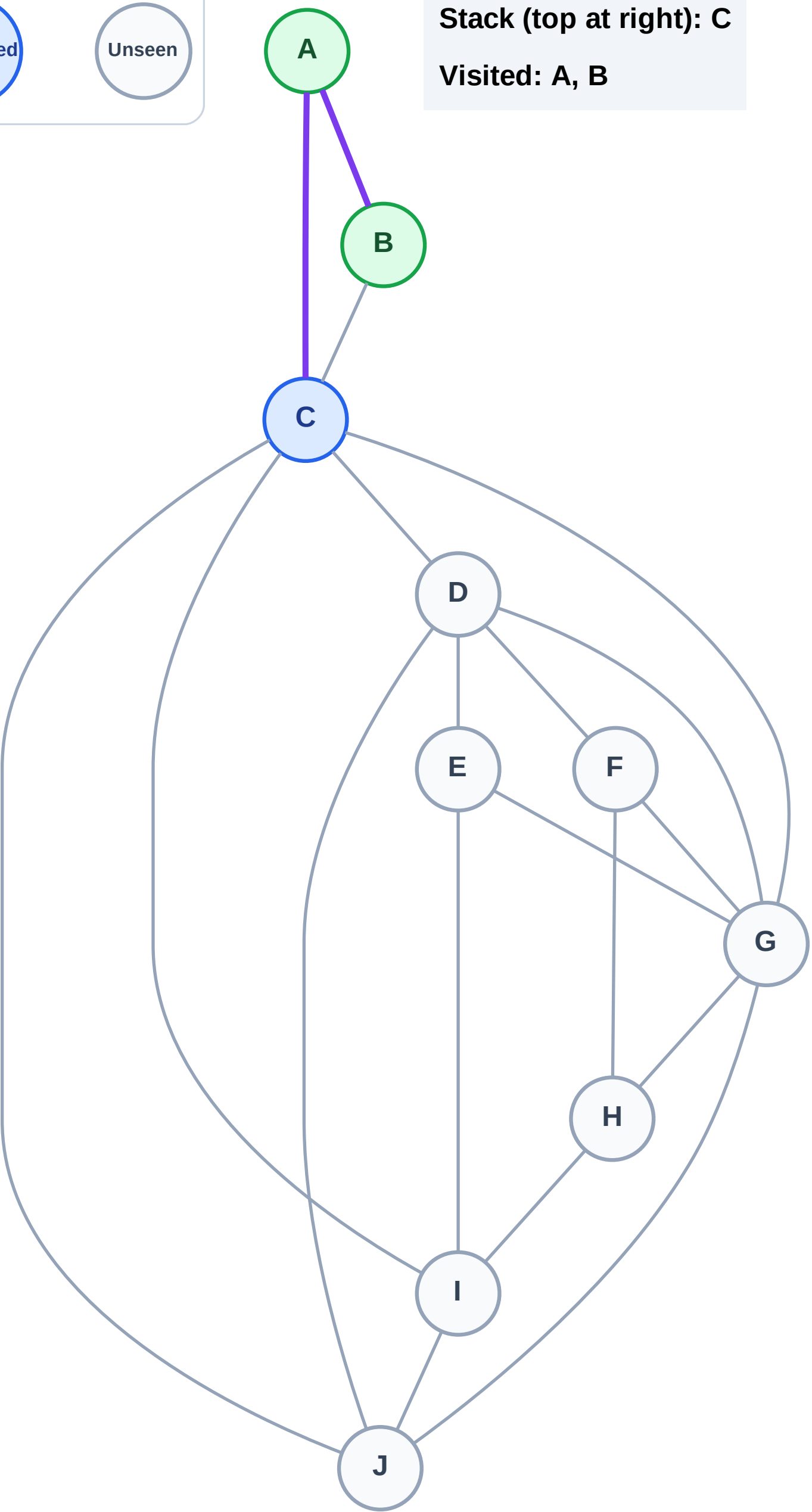
Processing

Discovered

Unseen

Stack (top at right): C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

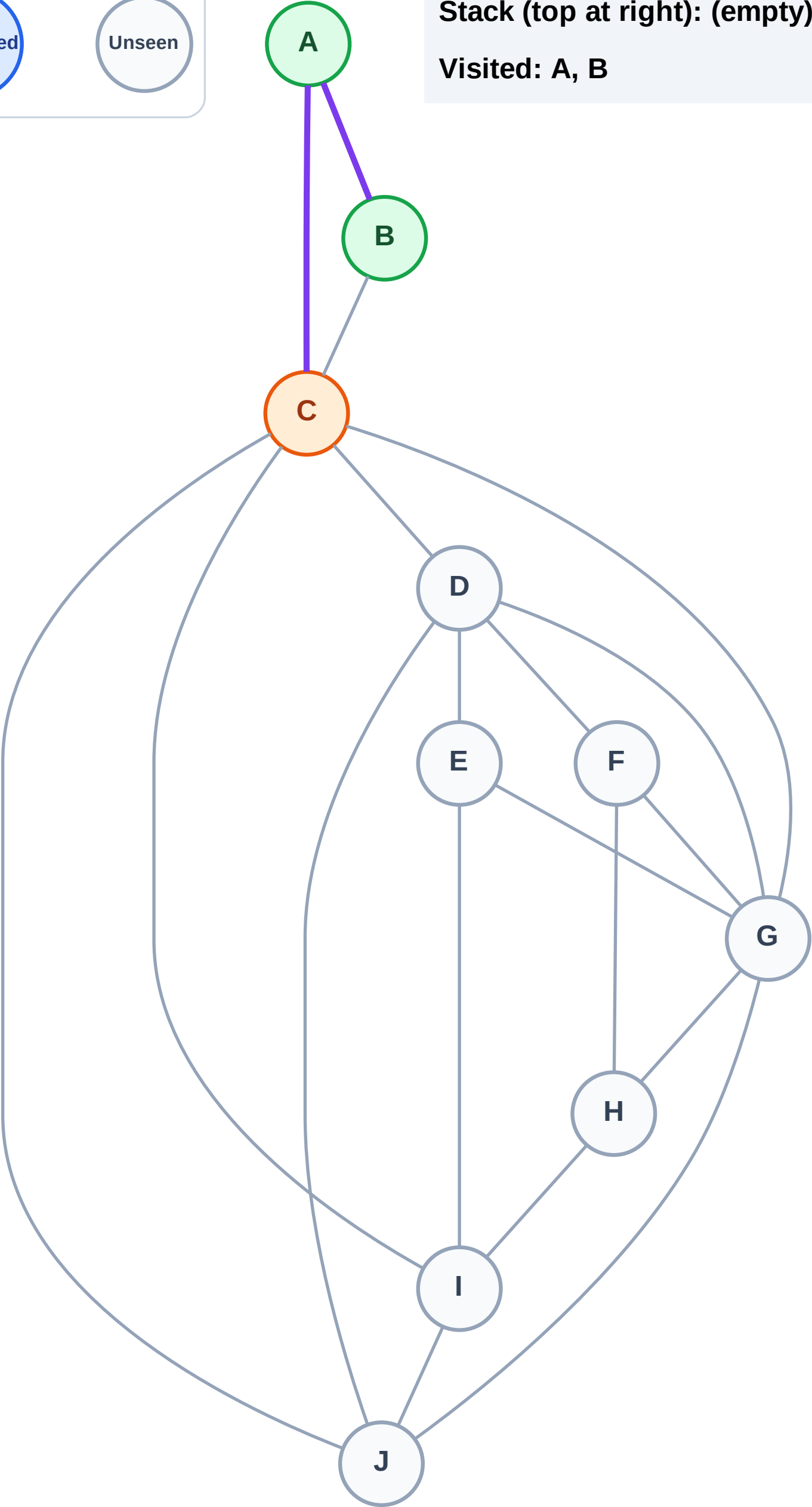
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B



DFS Traversal

Step 7: Discover J, I, G, D from C

Legend

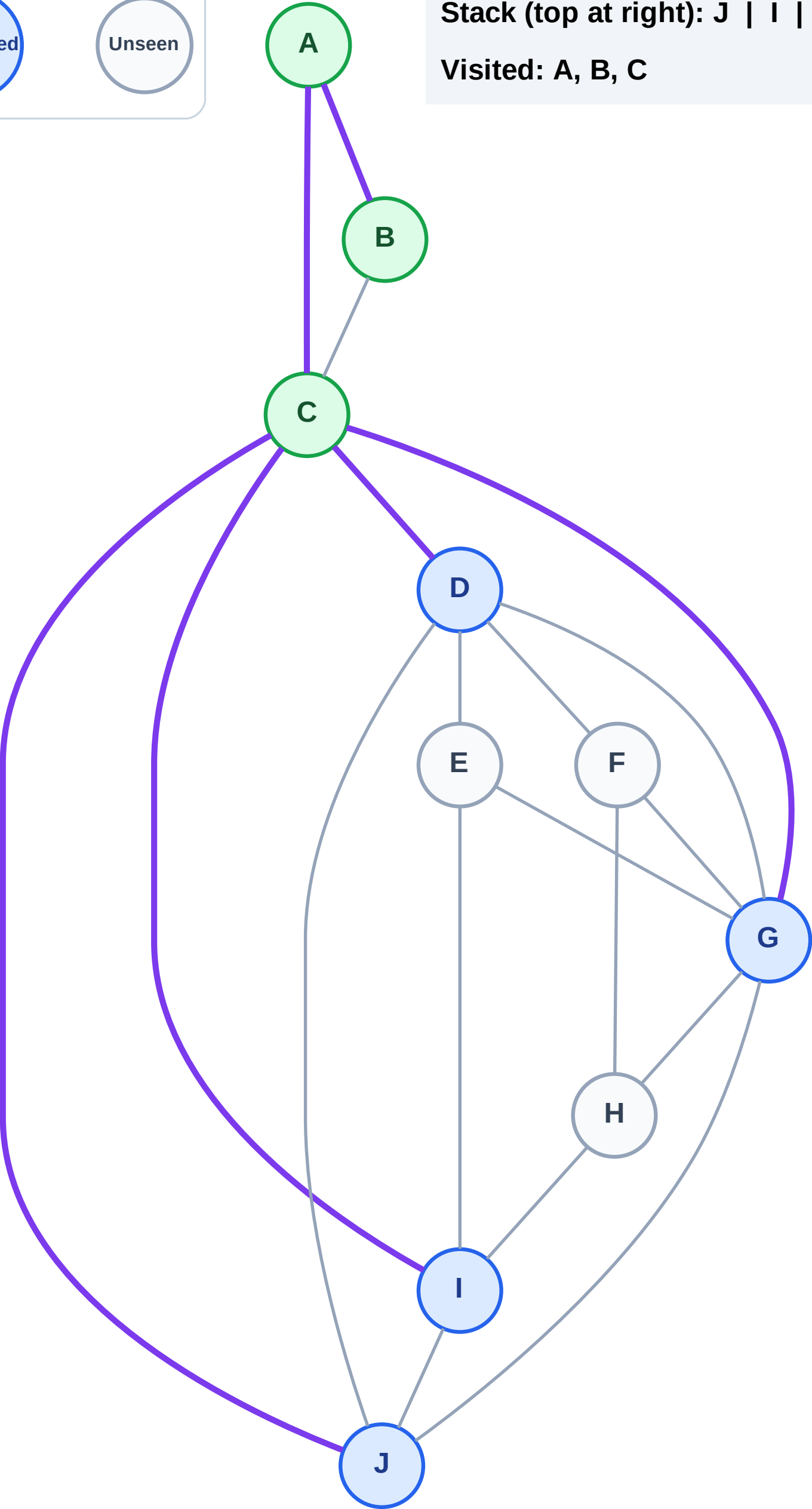
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | G | D
Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

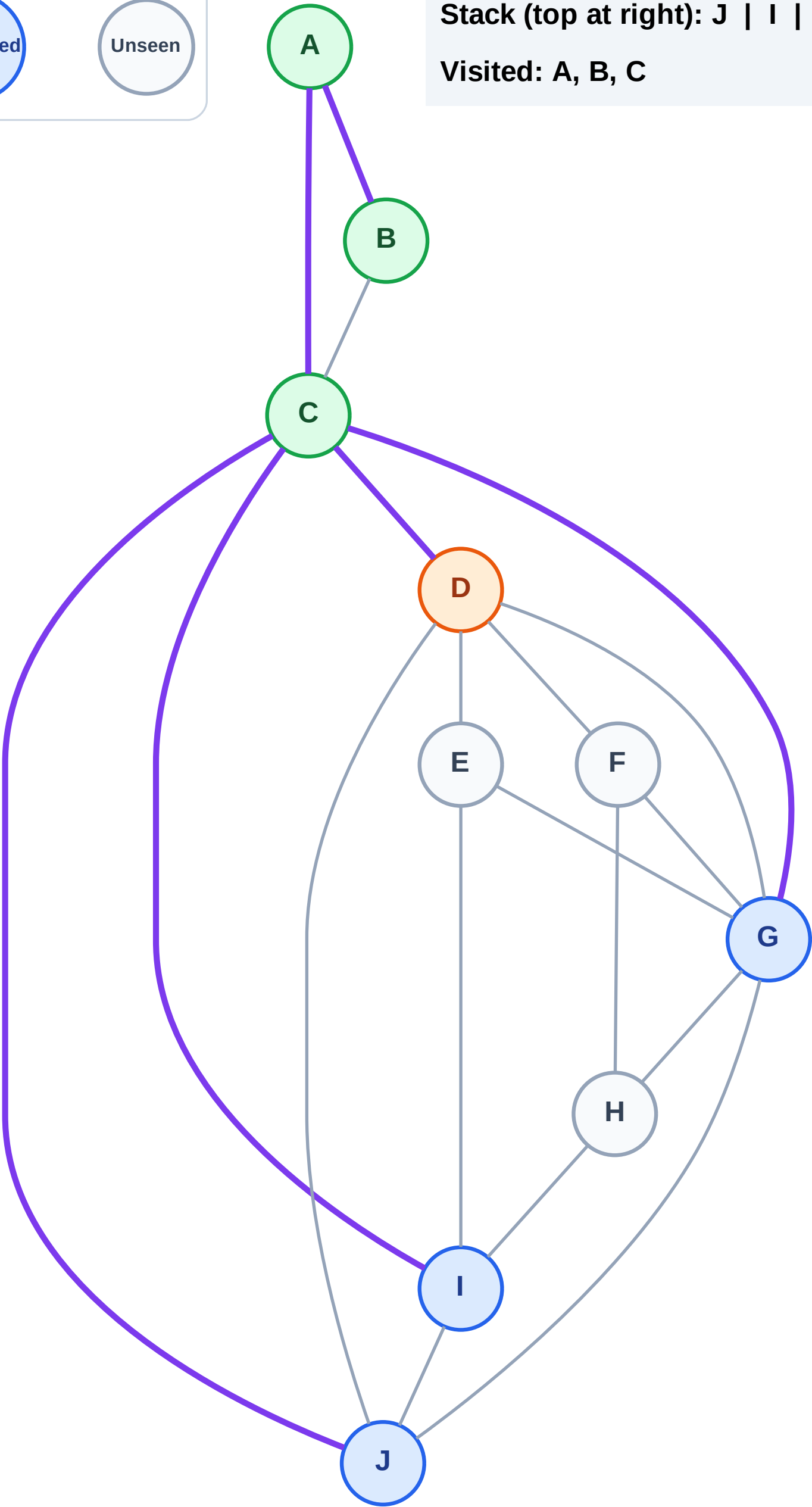
Processing

Discovered

Unseen

Stack (top at right): J | I | G

Visited: A, B, C



DFS Traversal

Step 9: Discover F, E from D

Legend

Complete

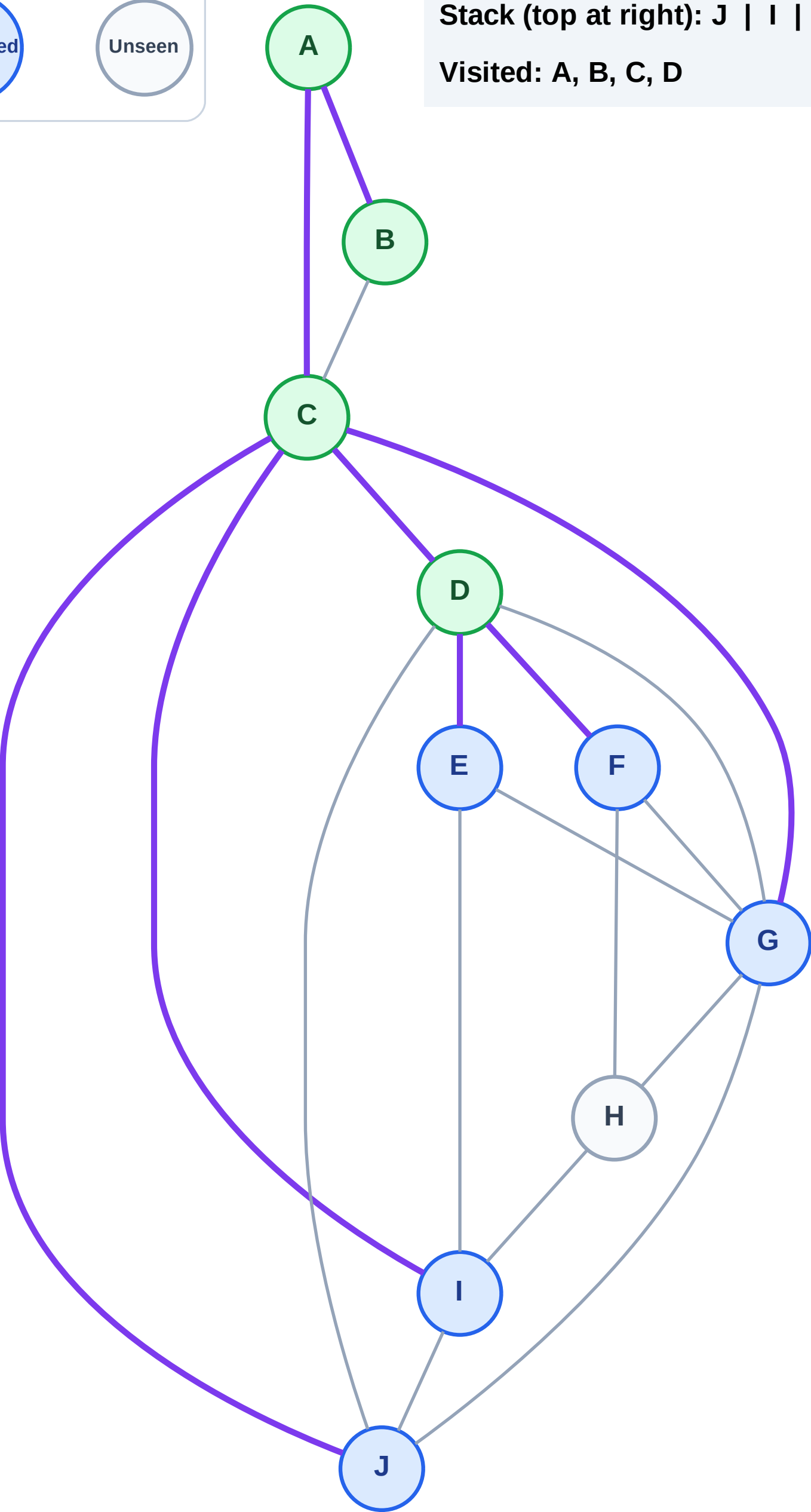
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | E

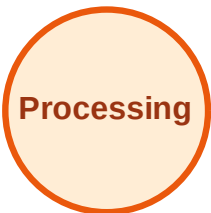
Visited: A, B, C, D



DFS Traversal

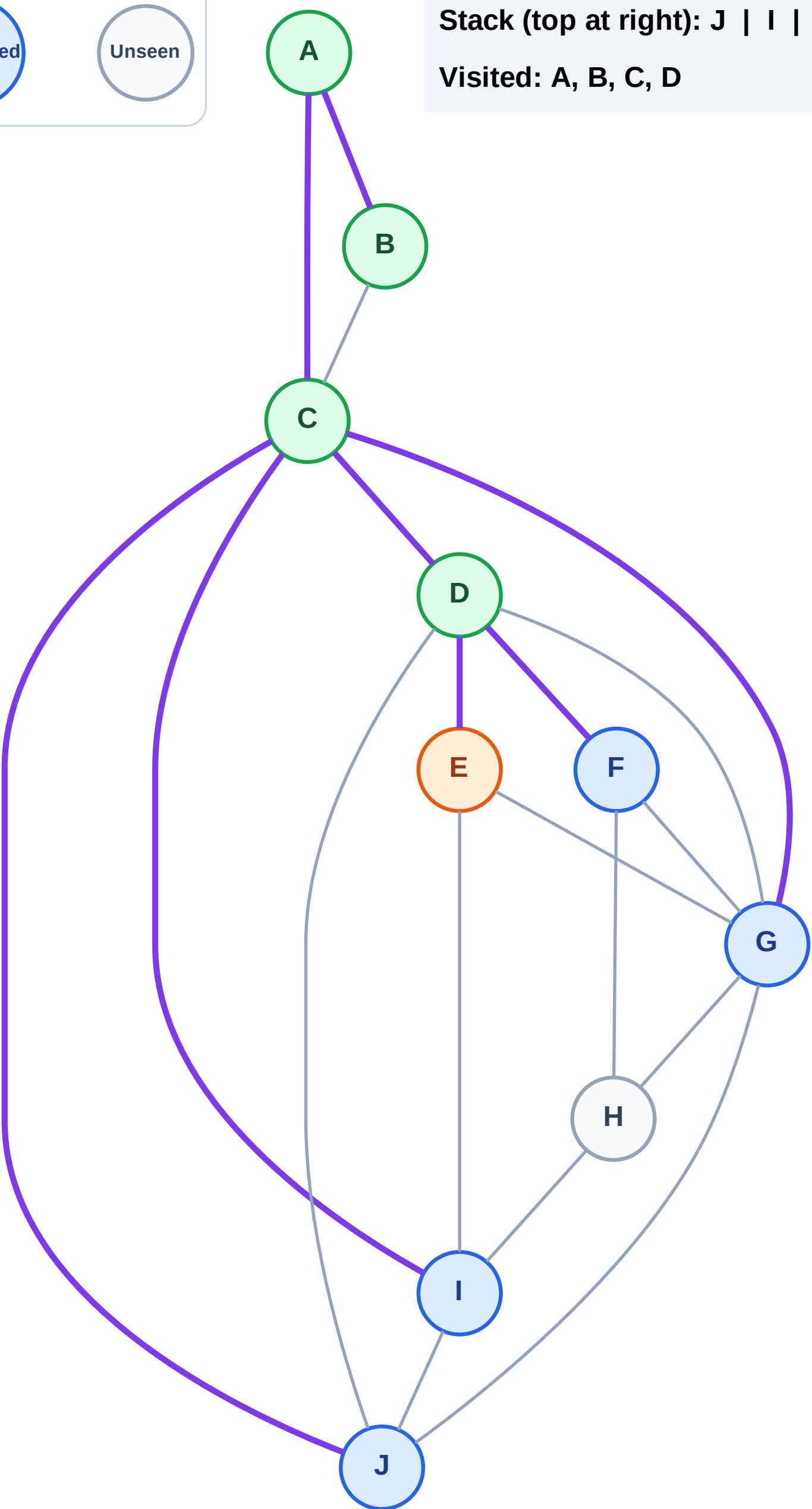
Step 10: Process node E

Legend



Stack (top at right): J | I | G | F

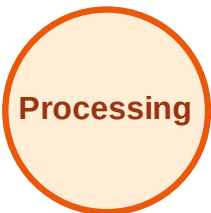
Visited: A, B, C, D



DFS Traversal

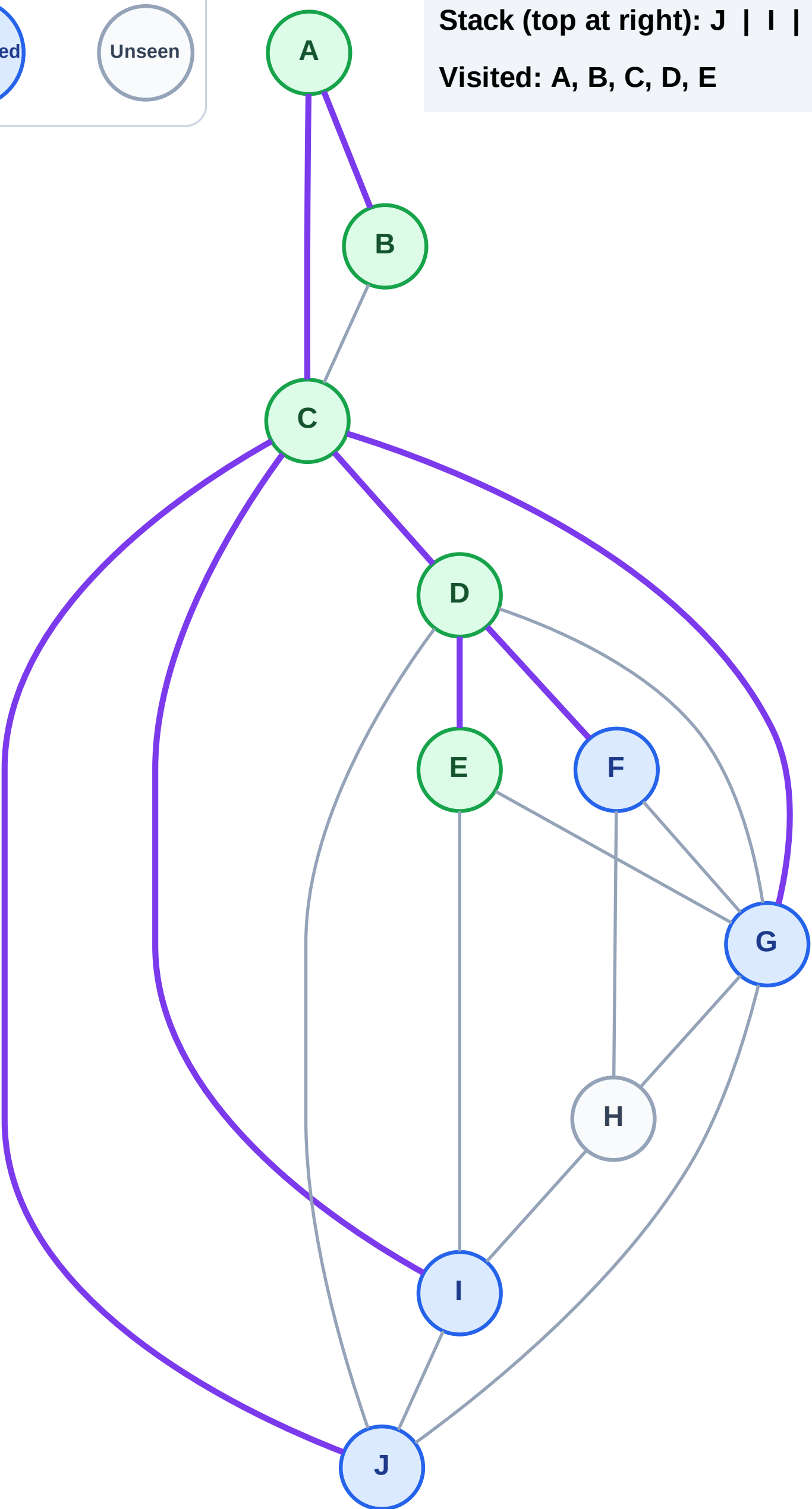
Step 11: No new neighbors from E

Legend



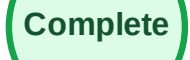
Stack (top at right): J | I | G | F

Visited: A, B, C, D, E

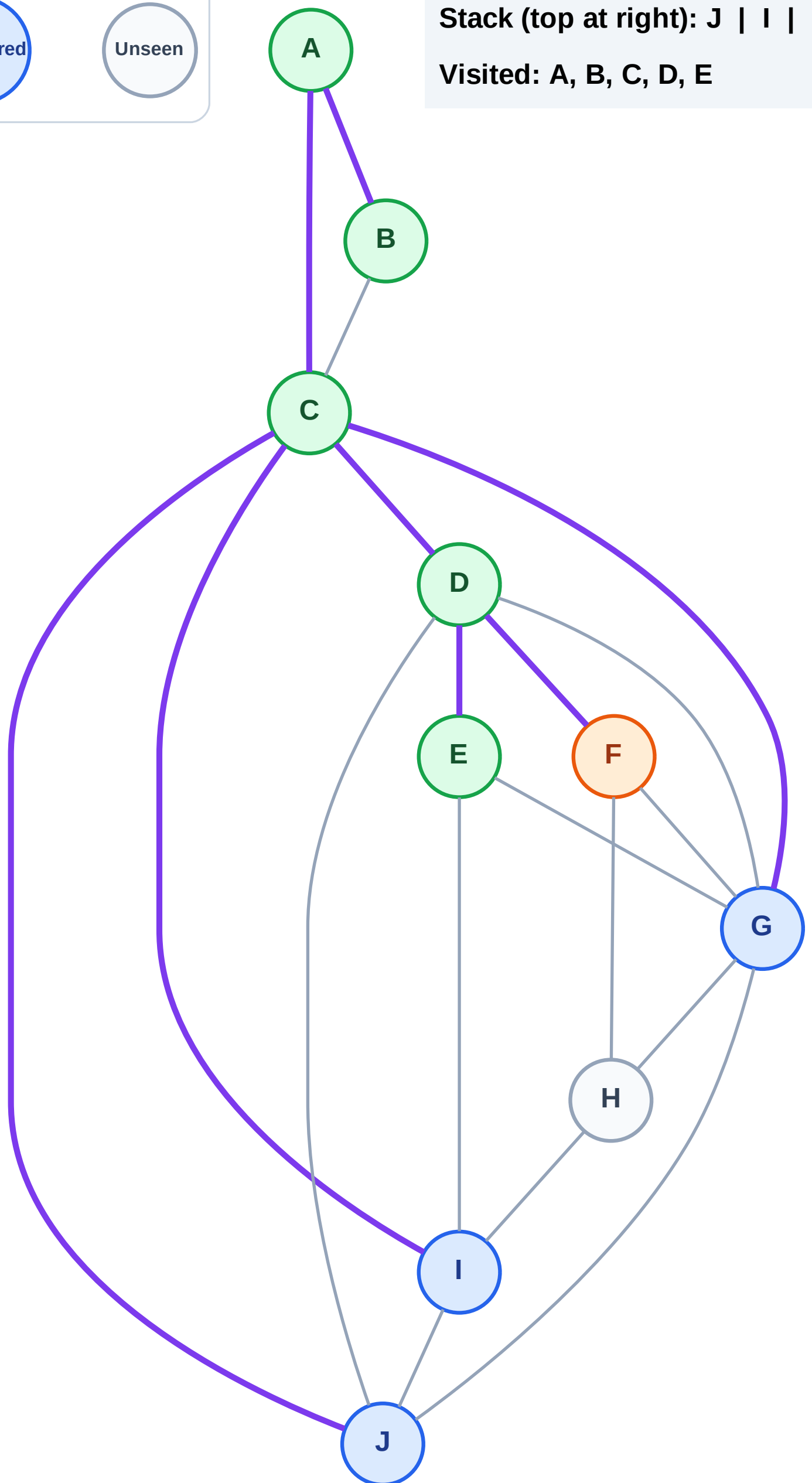


Step 12: Process node F

Page 10 of 10



Visited: A B C D E



DFS Traversal

Step 13: Discover H from F

Legend

Complete

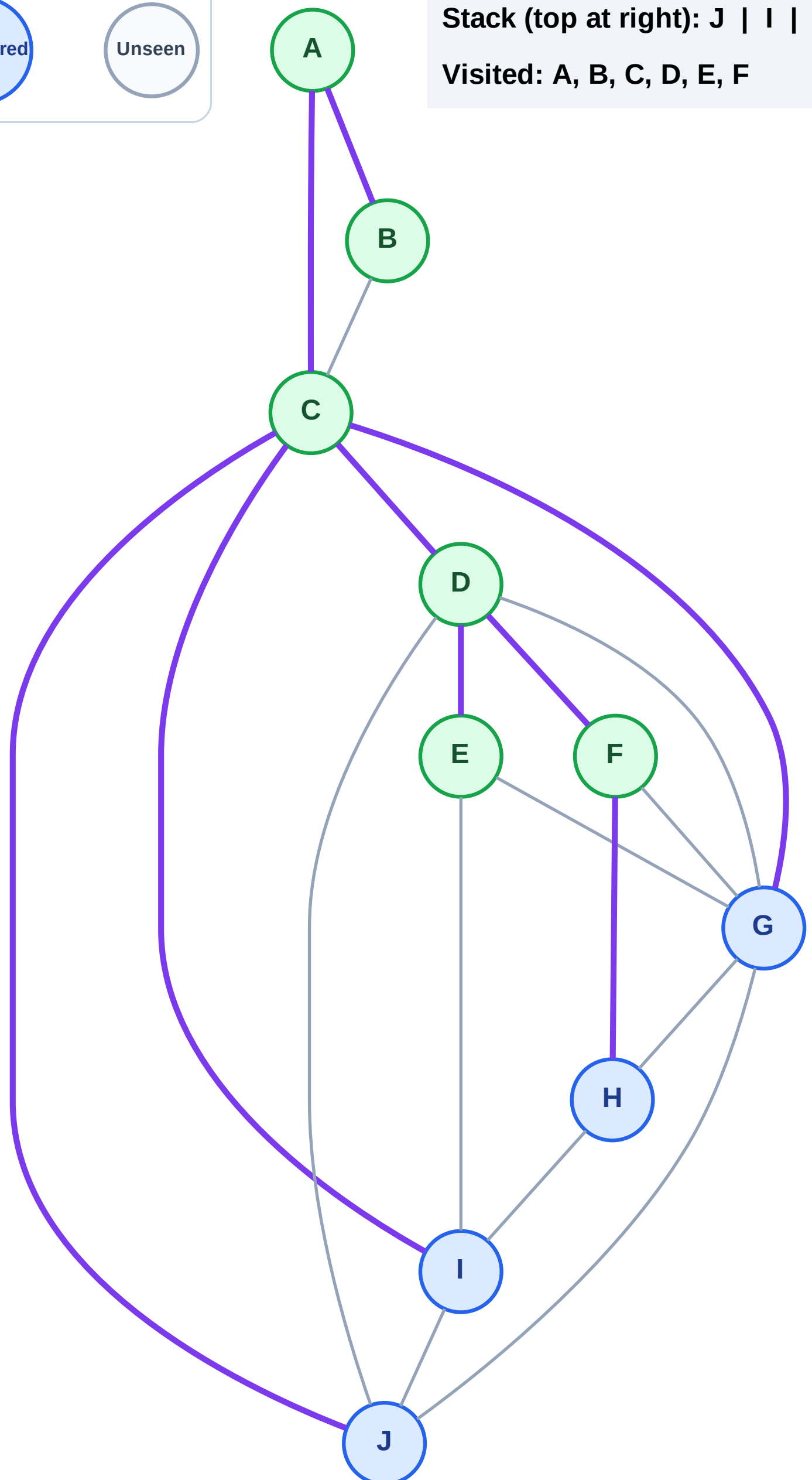
Processing

Discovered

Unseen

Stack (top at right): J | I | G | H

Visited: A, B, C, D, E, F



DFS Traversal

Step 14: Process node H

Legend

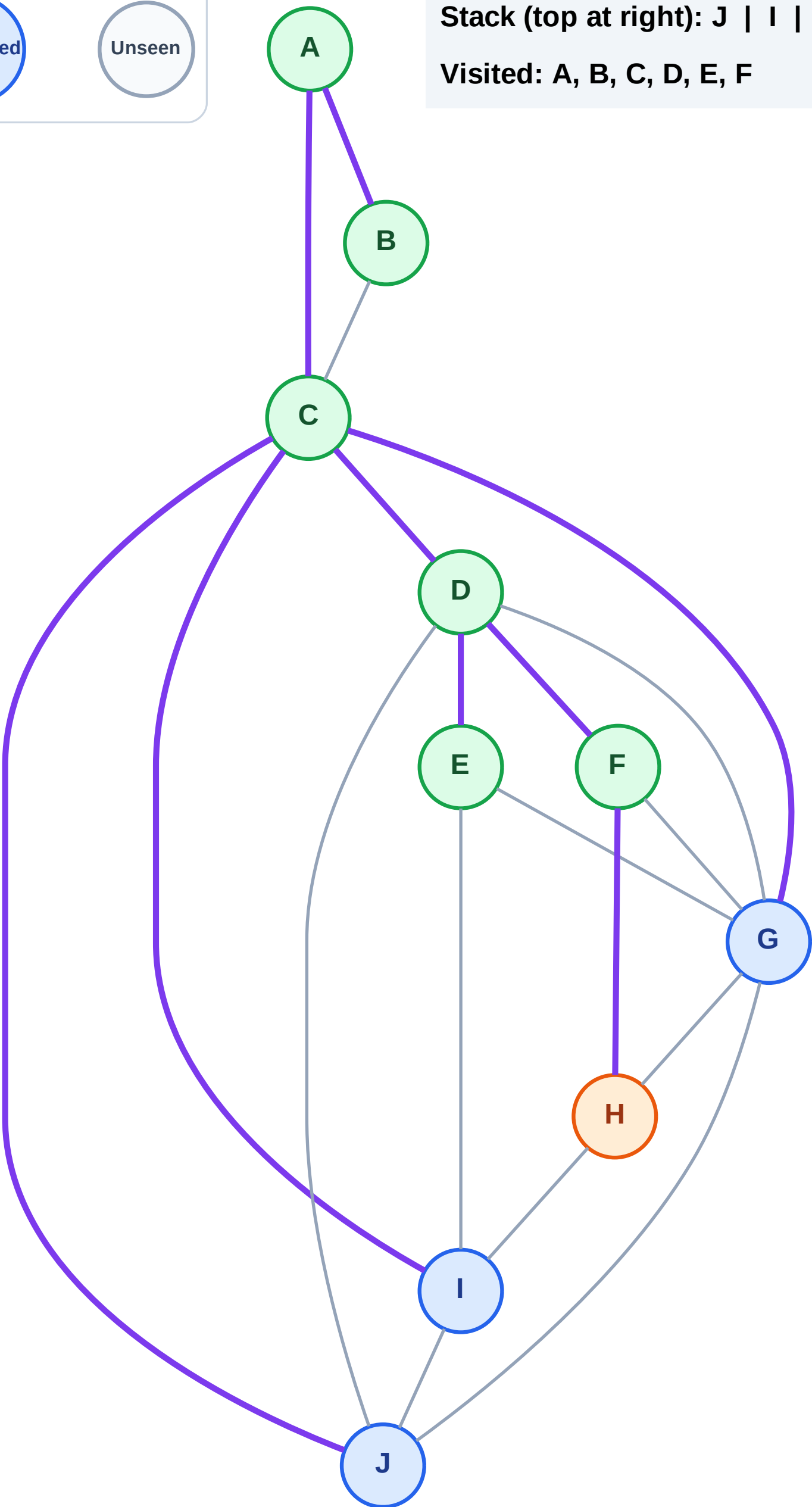
Complete

Processing

Discovered

Unseen

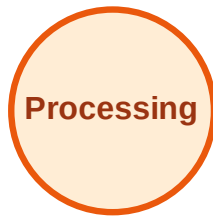
Stack (top at right): J | I | G
Visited: A, B, C, D, E, F



DFS Traversal

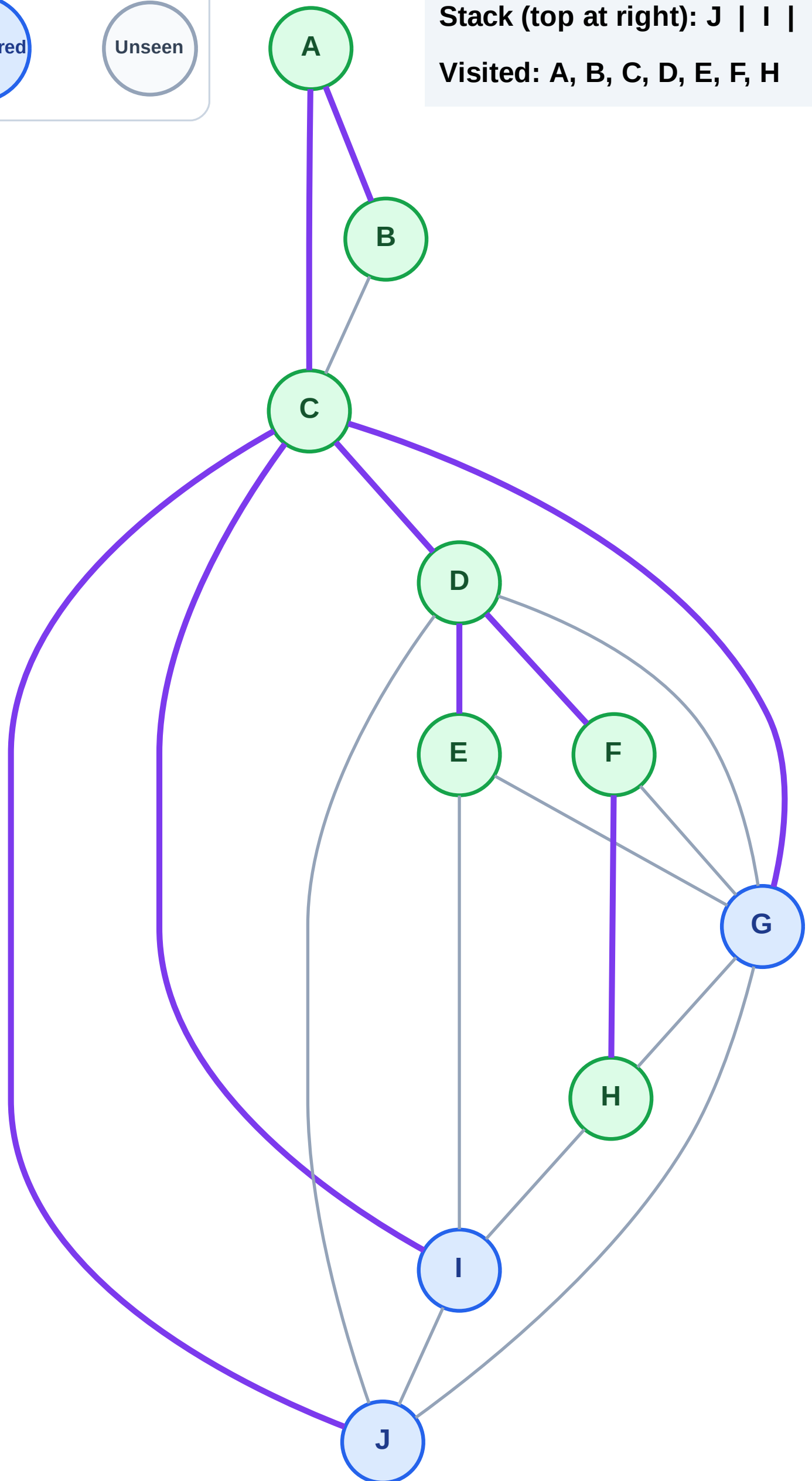
Step 15: No new neighbors from H

Legend



Stack (top at right): J | I | G

Visited: A, B, C, D, E, F, H



DFS Traversal

Step 16: Process node G

Legend

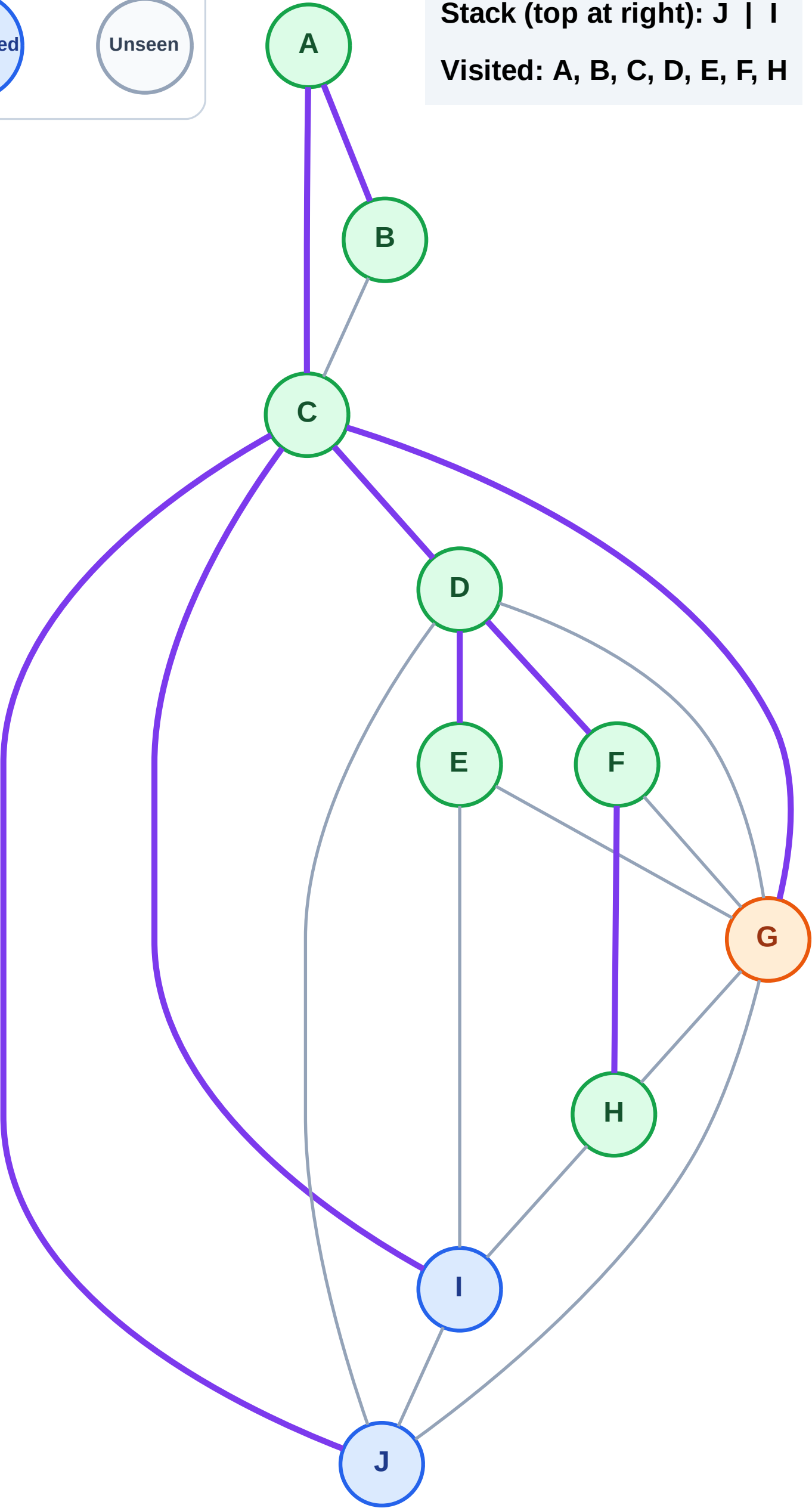
Complete

Processing

Discovered

Unseen

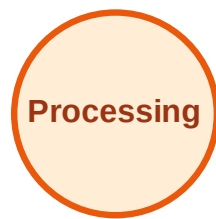
Stack (top at right): J | I
Visited: A, B, C, D, E, F, H



DFS Traversal

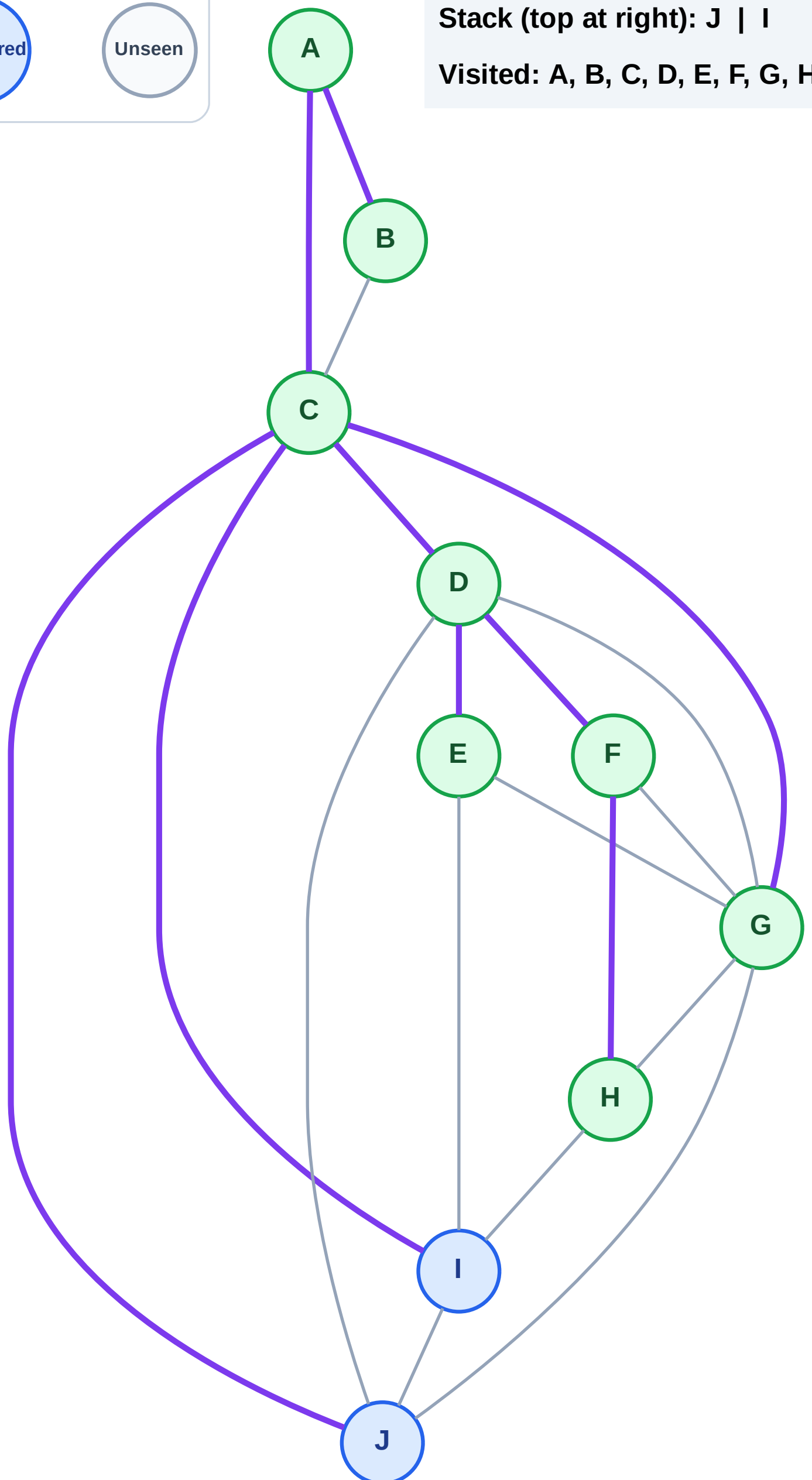
Step 17: No new neighbors from G

Legend



Stack (top at right): J | I

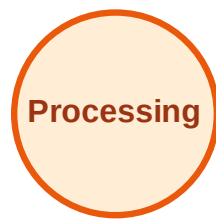
Visited: A, B, C, D, E, F, G, H



DFS Traversal

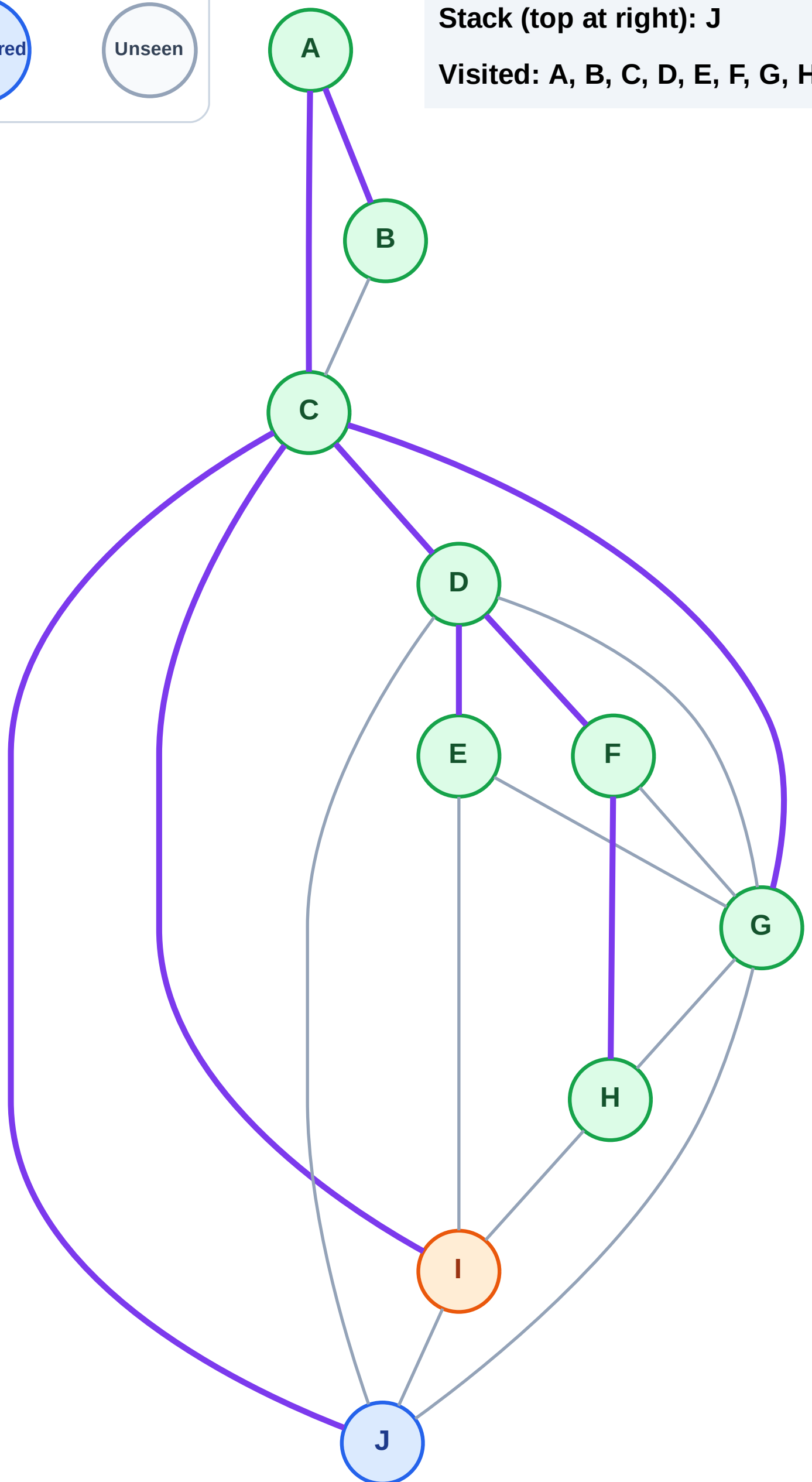
Step 18: Process node I

Legend



Stack (top at right): J

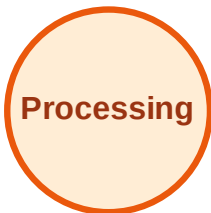
Visited: A, B, C, D, E, F, G, H



DFS Traversal

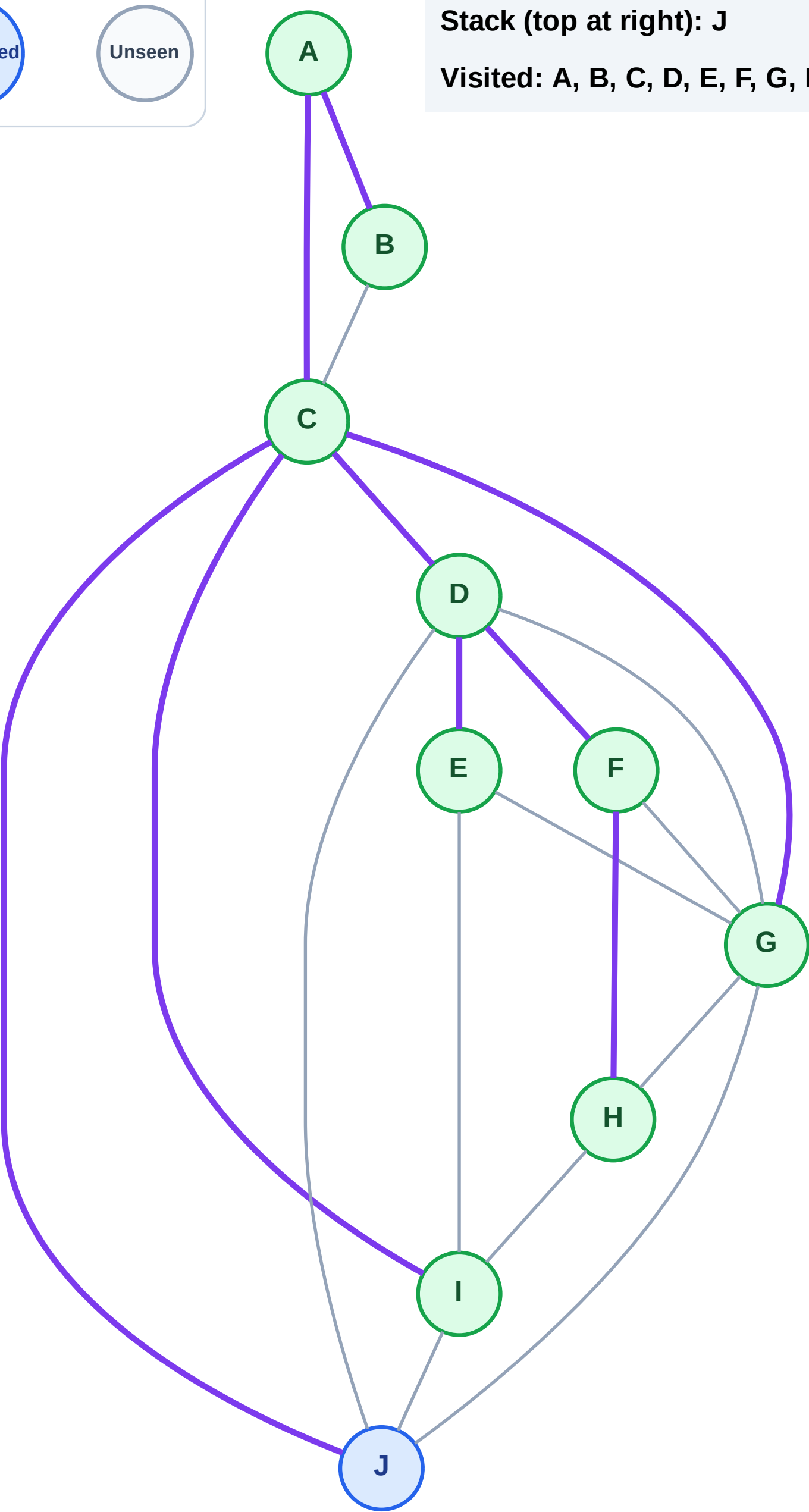
Step 19: No new neighbors from I

Legend



Stack (top at right): J

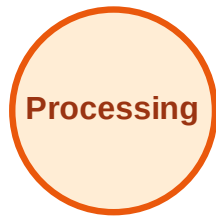
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

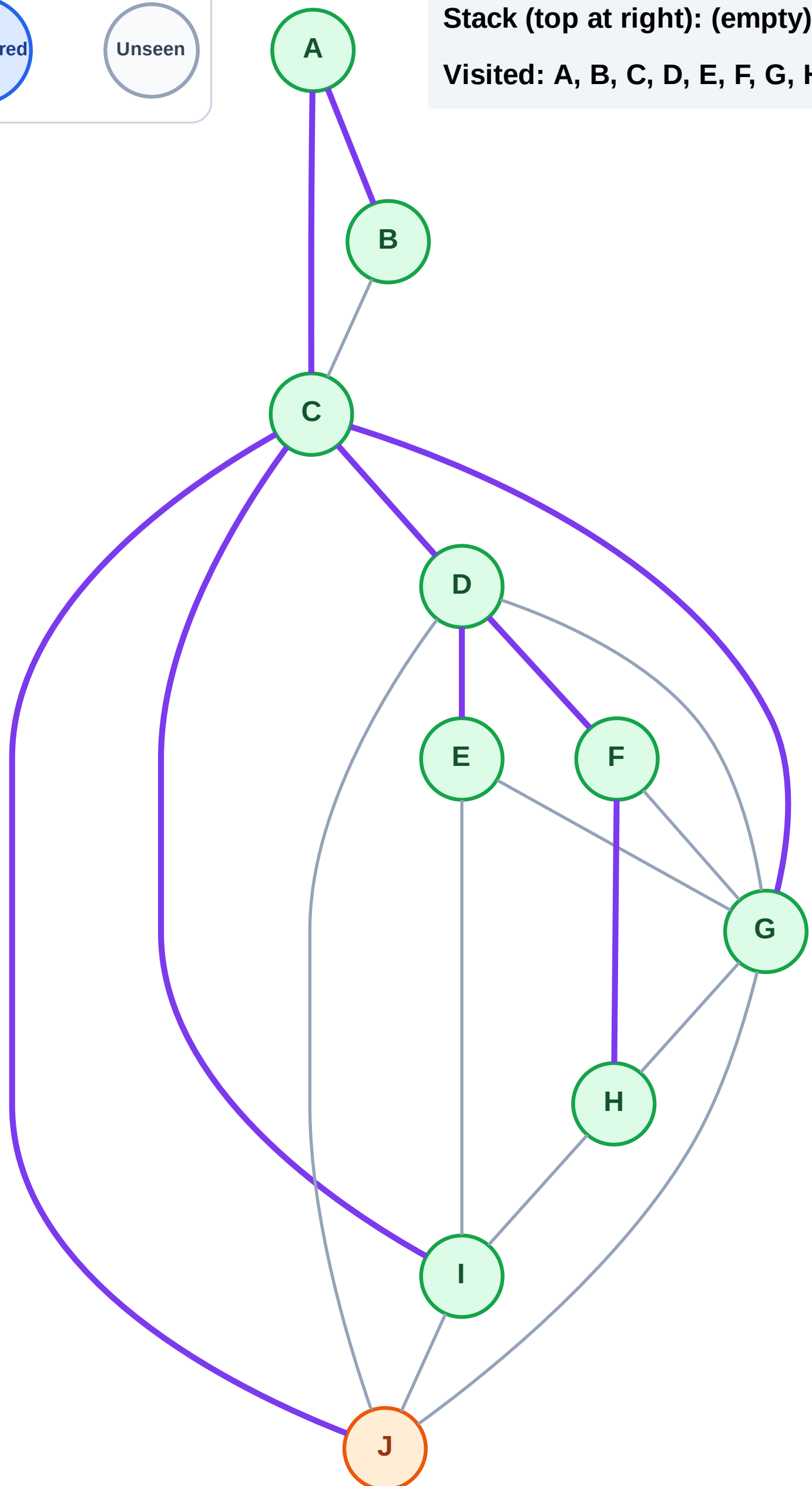
Step 20: Process node J

Legend



Stack (top at right): (empty)

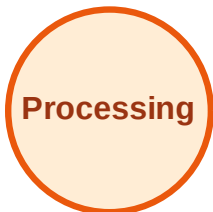
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

